

Oxide Perovskite Data Study Results of Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests

Somjit Roy^{1,†}, Pritam Dey¹, Bani K. Mallick¹ and Debdeep Pati²

¹Department of Statistics, Texas A&M University, College Station, TX 77843, U.S.A.

²Department of Statistics, University of Wisconsin-Madison, Madison, WI 53706, U.S.A.

[†]Corresponding Author: sroy_123@tamu.edu

Table of contents

A	The Oxide Perovskite Data	3
A.1	Symbolic Regression for Perovskite Catalyst Discovery	3
A.2	Oxide Perovskite Data Description	3
B	Experimental Settings of HierBOSSS and Competitors	6
B.1	HIERBOSSS	6
B.2	DSR	9
B.3	QLattice	10
B.4	SISSO++	11
B.5	gplearn	12
B.6	operon	13
B.7	PySR	14
B.8	BMS	16
B.9	BSR	17
C	Results of Competitors for the Oxide Perovskite Data Study	19
C.1	Results of DSR	19
C.2	Results of QLattice	20
C.3	Results of SISSO++	21
C.4	Results of gplearn	22
C.5	Results of operon	23
C.6	Results of PySR	24

C.7	Results of BMS	25
C.8	Results of BSR	27
D	Ablation Study for HierBOSSS: Practical Guidance for Choosing the Nominal Symbolic Forest Size	30
E	Results of HierBOSSS for the Oxide Perovskite Data Study	32
E.1	Descriptors Learned by HierBOSSS	32
E.2	Top Descriptors Ranked by JMP	33
E.3	Prediction Plots of V_{RHE} for HierBOSSS	39
E.4	Posterior Diagnostics for HierBOSSS	40
E.5	Runtime Comparison of HierBOSSS with the Bayesian SR Methods	40
F	Accuracy-Complexity Trade-off of Learned Descriptors	41

A The Oxide Perovskite Data

A.1 Symbolic Regression for Perovskite Catalyst Discovery

We consider the experimental oxide perovskite dataset of [Weng et al. \[2020\]](#), developed to identify compositional, electronic, and structural determinants of catalytic activity in the oxygen evolution reaction (OER). Oxide perovskites are a compositionally versatile family of materials with the general chemical form ABO_3 . The larger A -site cation primarily influences lattice geometry and structural stability, whereas the transition metal B -site cation governs much of the electronic structure and surface chemistry underlying catalytic activity [[Hwang et al., 2017](#)]. The ability to vary the chemical identities and mixing fractions at both crystallographic sites makes oxide perovskites important candidates for oxygen electrocatalysis, water splitting, renewable hydrogen production, and rechargeable metal-air batteries [[Suntivich et al., 2011](#), [Hwang et al., 2017](#)]. A central materials discovery problem is therefore to identify compact structure-activity relationships that explain OER performance and can guide the design of previously unexplored catalysts.

SR is particularly applicable to this problem as it searches directly for low-dimensional analytical expressions (materials genes or descriptors) relating catalytic activity to chemical, electronic, and geometric features [[Ghiringhelli et al., 2017](#), [Ouyang et al., 2018](#), [Wang et al., 2024](#)]. Such expressions can serve as interpretable descriptors rather than merely predictive surrogates [[Ghiringhelli et al., 2015](#), [Foppa et al., 2021](#)]. This role was demonstrated by [Weng et al. \[2020\]](#), where SR was not only implemented to identify the descriptor μ/t , combining the octahedral factor μ and tolerance factor t , but also used for screening new oxide perovskite compositions with improved OER activity. Related studies have likewise used SR and compressed sensing-based descriptor construction to recover compact physical relations in materials and catalytic systems [[Bartel et al., 2019](#), [Han et al., 2021](#)]. These developments motivate the use of **HIERBOSSS**’s complete probabilistic framework to recover interpretable catalytic descriptors. Unlike the `gplearn`-based [[Stephens, 2016](#)] analysis of [Weng et al. \[2020\]](#), which selected μ/t as a single Pareto-optimal descriptor, **HIERBOSSS** quantifies uncertainty across competing symbolic structure–activity relations, providing a more strong basis for guiding the design of new oxide perovskites.

A.2 Oxide Perovskite Data Description

We analyze the experimental dataset of [Weng et al. \[2020\]](#), comprising 18 conventional oxide perovskites synthesized and characterized under a common experimental protocol. The response variable is the potential V_{RHE} (in electron volts (eV)), measured relative to the reversible hydrogen electrode, required for a catalyst to sustain a prescribed current density (ρ). At a fixed ρ , a smaller V_{RHE} indicates that less applied potential is needed to drive the oxygen evolution reaction and therefore corresponds to greater catalytic activity.

For each material, 4 independently prepared samples were measured 3 times at each of 5 ρ levels: 0.05, 5, 10, 15, and 20 mA/cm². This yields $n = 18 \times 4 \times 3 \times 5 = 1080$ observations. The resulting activity landscape is shown in [Figure 1](#), where measurements are organized by material, experimental replicate, and ρ level. The landscape shows clear variation in V_{RHE}

across both material compositions and operating conditions, together with noticeable within-material experimental variability. Larger ρ values generally require larger applied potentials, while differences across materials reflect variation in intrinsic catalytic activity.

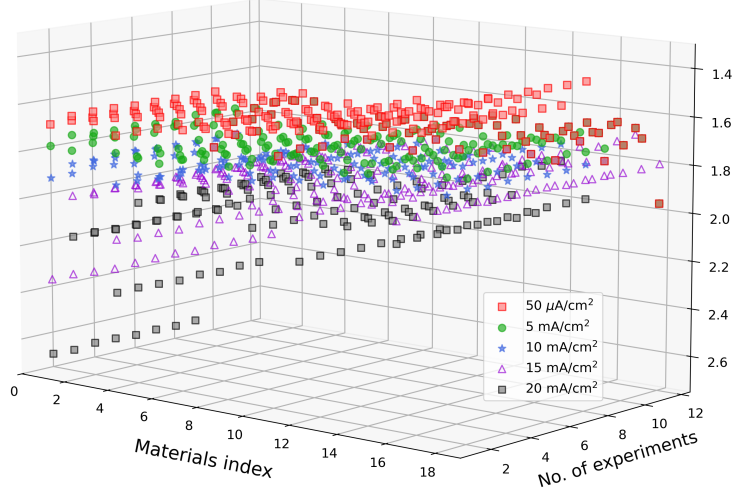


Figure 1: Experimental V_{RHE} measurements for the 18 oxide perovskites across 12 experimental replicates and 5 prescribed current density (ρ) levels.

Each observation is associated with the prescribed ρ and 7 chemically motivated materials descriptors: R_A , Q_A , N_d , χ_A , χ_B , μ , and t . Here, R_A denotes the composition-weighted ionic radius of the A -site cations, Q_A their composition-weighted formal charge, N_d the average number of f electrons associated with the transition metal B -site cations, and χ_A, χ_B are the composition-weighted electronegativities of the A - and B -site cations, respectively. When multiple elements occupy the same crystallographic site, the corresponding properties are averaged according to their stoichiometric fractions. For example, for a mixed A -site composition $A_x^1 A_{1-x}^2$, the effective ionic radius is $R_A = xR_{A^1} + (1-x)R_{A^2}$, with analogous weighting used for the other composition-dependent quantities.

The remaining two descriptors characterize the geometry of the perovskite lattice. The Goldschmidt tolerance factor, $t = (R_A + R_O)/[\sqrt{2}(R_B + R_O)]$, measures the compatibility of the constituent ionic sizes with the ideal perovskite lattice, where R_B is the composition-weighted ionic radius of the B -site cations and R_O is the ionic radius of O^{2-} . It is commonly used as a first-order descriptor of perovskite formability and structural distortion. the octahedral factor, $\mu = R_B/R_O$, measures the size compatibility of the B -site cation with the surrounding BO_6 oxygen octahedron. The resulting SR problem therefore has response V_{RHE} and $p = 8$ predictors, i.e., $(n, p) = (1080, 8)$; a concise summary is provided in Table 1.

Table 1: Response and input variables in the oxide perovskite data study.

Role	Variable	Description	Scientific interpretation
Output	V_{RHE}	Potential relative to the reversible hydrogen electrode required to sustain a prescribed current density.	Measures OER performance. At a fixed current density, a smaller value indicates a smaller required applied potential and hence greater catalytic activity.

Continued on next page

Table 1: Response and input variables in the oxide perovskite data study (*continued*).

Role	Variable	Description	Scientific interpretation
Input	ρ	Prescribed current density, taking values 0.05, 5, 10, 15, and 20 mA/cm ² .	Defines the electrochemical operating condition at which V_{RHE} is measured. Larger current densities generally require greater applied potential to sustain a higher reaction rate.
Input	R_A	Composition-weighted ionic radius of the A -site cations, measured in Å.	Characterizes the effective size of the cations occupying the larger crystallographic site and influences lattice expansion, octahedral tilting, structural distortion, and the tolerance factor t .
Input	Q_A	Composition-weighted formal charge of the A -site cations.	Contributes to charge neutrality and indirectly constrains the oxidation states and electronic configurations of the catalytically active B -site transition metals.
Input	N_d	Composition-weighted number of d electrons associated with the transition metal B -site cations.	Summarizes the B -site electronic configuration, which influences metal–oxygen covalency, orbital occupancy, adsorption energetics, and electron transfer during the OER.
Input	χ_A	Composition-weighted electronegativity of the A -site cations.	Describes the tendency of the A -site species to attract electronic charge and can indirectly affect bonding, charge redistribution, and B -site oxidation states.
Input	χ_B	Composition-weighted electronegativity of the transition metal B -site cations.	Reflects the electronic character of the catalytically active site and its interaction with oxygen ligands and adsorbed OER intermediates.
Input	t	Goldschmidt tolerance factor, $t = (R_A + R_O)/[\sqrt{2}(R_B + R_O)]$.	Measures ionic size compatibility with the ideal ABO_3 lattice. Deviations from the ideal geometric range are associated with octahedral rotation, lattice distortion, and changes in structural stability.
Input	μ	Octahedral factor, $\mu = R_B/R_O$.	Measures the size compatibility of the B -site cation with its surrounding oxygen octahedron and is related to the formation and distortion of the BO_6 units central to perovskite structure and catalysis.

B Experimental Settings of **HierBOSSS** and Competitors

Computational environment. All experiments related to the oxide perovskite dataset were implemented in Python and executed on a MacBook Air equipped with an Apple M2 processor and 8GB of unified memory.

B.1 **HierBOSSS**

We evaluate **HierBOSSS** for discovering symbolic descriptors of catalytic activity from the oxide perovskite dataset of Weng et al. [2020]. We perform 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, **HierBOSSS** is fitted to the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $\text{MAE} = n^{-1} \sum_{i=1}^n |V_{\text{RHE},i} - \hat{V}_{\text{RHE},i}|$.

HierBOSSS data configuration for the oxide perovskite study

```

1 perovskite_data_settings = {
2     "response": "VRHE",
3     "features": [
4         "RA",          # R_A
5         "QA",          # Q_A
6         "Nd",          # N_d
7         "XA",          # chi_A
8         "XB",          # chi_B
9         "mu",          # octahedral factor
10        "t",            # tolerance factor
11        "current_density", # rho
12    ],
13    "n_observations": 1080,
14    "split_seeds": ...,
15    "n_splits": 5,
16    "train_size": 0.90,
17    "test_size": 0.10,
18 }
```

The operator set $\mathbb{O}$ is kept fixed across all 5 data splits and matched with the operators used in Weng et al. [2020].

HierBOSSS operator set

```

1 opset = [
2     "add",    # (x, y) -> x + y
3     "mul",    # (x, y) -> x*y
4     "neg",    # x -> -x
5     "inv",    # x -> 1/x, protected near zero
6     "sqrt",   # x -> sqrt(abs(x))
7 ]
```

The prior hyperparameter specifications are shared across all data splits. The Dirichlet concentration parameters for operator and feature assignments are set to one, i.e., $\alpha_{\text{op}} = (1, \dots, 1)$

and $\alpha_{ft} = (1, \dots, 1)$. The depth-dependent splitting probability is controlled by $(\alpha_0, \delta_0) = (0.95, 2.00)$. For the outer regression coefficient vector β , including an intercept, we use prior mean $\mu_\beta = \mathbf{0}$ and covariance $\Sigma_\beta = 10\mathbf{I}_{K+1}$. The model noise variance σ^2 is assigned an Inverse-Gamma prior with hyperparameters $\nu = 0.05 = \lambda$.

HierBOSSS prior configuration

```

1 HierBOSSS_prior = {
2     # Dirichlet concentration parameters
3     "alpha_op": "ones(len(opset))",
4     "alpha_ft": "ones(number_of_features)",
5
6     # depth-dependent split probability
7     "alpha_0": 0.95,
8     "delta_0": 2.00,
9
10    # Gaussian prior for the outer regression coefficients
11    "add_intercept": True,
12    "beta0": "zeros(K + 1)",
13    "V0": "10 * I_(K + 1)",
14
15    # Inverse-Gamma prior for the model noise variance
16    "a0": 0.05,
17    "b0": 0.05,
18    "sigma2_prior": "Inverse-Gamma(a0 / 2, b0 / 2)",
19 }
```

We set the nominal symbolic forest size to $K = 8$. The choice $K = 8$ is guided by the ablation study in Section D, which shows that predictive performance stabilizes around this value while computational cost continues to increase for larger symbolic forests. Therefore, we use the corresponding ablation study as a practical guideline for choosing the nominal value of K . Each MCMC chain is run for 2000 iterations.

HierBOSSS symbolic forest and MCMC configuration

```

1 HierBOSSS_perovskite_settings = {
2     "K": 8,                # nominal symbolic forest size
3     "maxiter": 2000,      # MCMC iterations within each chain
4 }
```

For each train/test split of the oxide perovskite dataset, we run 5 independent parallel MCMC chains of HierBOSSS. The sampled symbolic forests from all chains are subsequently pooled and ranked according to their joint marginal posterior over $\mathcal{T}$, i.e., $\text{JMP}(\mathcal{T})$.

HierBOSSS parallel chain configuration

```

1 parallel_mcmc_settings = {
2     "nchains": 5,          # number of parallel MCMC chains
3     "burnin": 0,
4     "thin": 1,
5     "n_jobs": 5,
```

continued on next page

continued from previous page

```

6     "chain_seed_rule": "...",
7     "target": "JMP(T)",
8 }

```

The MH kernel uses seven local symbolic tree proposal moves: grow (g), prune (p), change feature (cf), change operator (co), subtree replacement (st), delete node (del), and insert node (ins). All available moves receive equal proposal weight through the move weight vector π_{move} . At each tree update, the sampler determines the moves and node addresses that are valid for the current tree, normalizes the proposal probabilities over this available set, and then samples a move and a valid address.

HierBOSSS local symbolic tree proposal moves

```

1 # local symbolic tree proposal move weights
2
3 move_weights = {
4     "grow": 1.0,           # g
5     "prune": 1.0,         # p
6     "change_feature": 1.0, # cf
7     "change_operator": 1.0, # co
8     "subtree_replace": 1.0, # st
9     "delete_node": 1.0,    # del
10    "insert_node": 1.0,     # ins
11 }

```

The initial symbolic forests and proposal-generated subtrees use uniform operator and feature weights, i.e., uniform choices for $(\mathbf{w}_{\text{op,init}}, \mathbf{w}_{\text{op,prop}}, \mathbf{w}_{\text{ft,init}}, \mathbf{w}_{\text{ft,prop}})$. These weights govern only initialization and proposal generation; the posterior target uses the Dirichlet-collapsed symbolic forest prior $\Pi_{\text{forest},K}(\mathcal{T} \mid \alpha_0, \delta_0, \alpha_{\text{op}}, \alpha_{\text{ft}})$. Thus, preferences for particular operators and material features are learned through their marginal prior contribution to the joint posterior rather than fixed through the proposal mechanism.

HierBOSSS initialization and proposal weights

```

1 # initialization of operator and feature weight vectors with uniform choice
2 # proposal operator and feature weight vectors with uniform choice
3
4 initialization_and_proposal = {
5     "wts_init_op": "ones(len(opset))",
6     "wts_init_ft": "ones(number_of_features)",
7     "wts_prop_op": "wts_init_op",
8     "wts_prop_ft": "wts_init_ft",
9     "proposal_normalization":
10         "normalize over available moves and valid labels",
11 }

```

After sampling, retained symbolic forests from the parallel MCMC chains are pooled and ranked by $\log \text{JMP}(\mathcal{T})$. For each train/test split of the oxide perovskite dataset, the top $r = 10$ symbolic forests are retained as the representative high-posterior set. For each retained forest,

the posterior mean of the outer regression coefficient vector is computed under the conjugate Gaussian layer, producing its corresponding symbolic forest expression (*Raw*).

HierBOSSS posterior ranking configuration and obtaining *Raw* expressions

```

1 posterior_ranking = {
2     "ranking_score": "log_JMP",
3     "pooling": "retained forests from all 5 parallel chains",
4     "top_r": 10,
5     "raw_expression_coefficients":
6         "posterior mean under the conjugate NIG layer",
7 }
```

Finally, each of the top-ranked symbolic forests is passed through the post-MCMC symbolic model refinement step in **HierBOSSS**. Conditional on a fixed forest, its K symbolic trees are treated as candidate additive components, and a smaller subset is selected by minimizing BIC. The outer regression coefficients are then re-estimated and the resulting expression is simplified using SymPy [Meurer et al., 2017]. The reported *Final* expression therefore has a data-adaptive effective forest size $K_{\text{eff}} \leq K$.

HierBOSSS post-MCMC symbolic model refinement to obtain *Final* expressions

```

1 post_mcmc_refinement = {
2     "candidate_models": "top_r forests ranked by log_JMP",
3     "candidate_components":
4         "symbolic trees within each fixed forest",
5     "criterion": "BIC",
6     "force_intercept": False,
7     "coefficient_refit":
8         "posterior mean on the selected symbolic design",
9     "simplification": "SymPy",
10    "effective_forest_size":
11        "K_eff = number of retained non-intercept trees",
12 }
```

B.2 DSR

We use the `dso.DeepSymbolicRegressor` implementation (<https://github.com/dso-org/deep-symbolic-optimization>) of Deep Symbolic Regression (DSR) [Petersen et al., 2021]. We consider 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, DSR is fitted to the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $\text{MAE} = n^{-1} \sum_{i=1}^n |V_{\text{RHE},i} - \hat{V}_{\text{RHE},i}|$.

Following Weng et al. [2020], the operator set is configured as follows.

DSR operator set

```

1 function_set = [
2     "add",      # (x, y) -> x + y
3     "sub",      # (x, y) -> x - y
4     "mul",      # (x, y) -> x*y
5     "div",      # (x, y) -> x/y
6     "sqrt",     # x -> sqrt(x)
7 ]

```

Protected operators are enabled throughout the analysis, so division and square root evaluations are numerically safeguarded during symbolic program search. The following DSR configuration is maintained across all 5 train/test splits.

DSR settings

```

1 dsr_settings = {
2     "n_samples": 200000, # total symbolic program sampling budget
3     "batch_size": 500,   # programs per RNN policy-gradient update
4     "max_length": 25,    # maximum sampled expression length
5 }

```

These settings correspond to approximately 400 RNN policy-gradient updates unless training terminates early. During training, DSR retains the highest-reward symbolic program encountered during the search. The final learned expression is extracted from the fitted estimator through

Extracting the final DSR expression

```

1 estimator.program_

```

which stores the best symbolic program returned by `dso.DeepSymbolicRegressor`.

B.3 QLattice

We use the `feyn.QLattice` implementation (<https://docs.abzu.ai/>) of QLattice [Broløs et al., 2021]. We consider 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, QLattice is fitted to the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $\text{MAE} = n^{-1} \sum_{i=1}^n |V_{\text{RHE},i} - \hat{V}_{\text{RHE},i}|$.

Following Weng et al. [2020], the primitive functions are chosen as follows.

QLattice operator set

```

1 function_names = [
2     "inverse", # z -> 1/z
3     "linear",  # z -> a*z + b
4     "sqrt",    # z -> sqrt(z)
5     "add",     # (z1, z2) -> z1 + z2

```

continued on next page

continued from previous page

```

6     "multiply", # (z1, z2) -> z1*z2
7 ]

```

For each train/test split, the following QLattice configuration is maintained.

QLattice settings

```

1 qlattice_settings = {
2     "n_epochs": 25,          # number of symbolic search epochs
3     "max_complexity": 20,    # maximum sampled model complexity
4     "criterion": "bic",      # model fitting and ranking criterion
5 }

```

After the final epoch, the first model in the BIC-ranked collection is retained as the learned expression. The recovered symbolic formula is extracted using SymPy [Meurer et al., 2017] through

Extracting the final QLattice expression

```

1 # signif -> significant digits in the fitted numerical coefficients
2 best_model = models[0]
3 expression = best_model.sympify(signif=8)

```

which converts the selected QLattice model into a symbolic expression.

B.4 SISSO++

For sure independence screening and sparsifying operator (SISSO) [Ouyang et al., 2018], we use the SISSORegressor implementation (<https://sissopp-developers.gitlab.io/sissopp/>) from SISSO++ [Purcell et al., 2022]. We consider 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, SISSO++ is fitted to the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $\text{MAE} = n^{-1} \sum_{i=1}^n |V_{\text{RHE},i} - \hat{V}_{\text{RHE},i}|$.

For all 5 train/test splits, the SISSO++ regression task is run with the following common configuration.

SISSO++ settings

```

1 sisso_settings = {
2     "n_dim": 3,              # number of selected symbolic descriptors
3     "max_rung": 3,           # maximum recursive feature-construction depth
4     "n_sis_select": 50,      # candidate descriptors retained by SIS
5     "n_residual": 20,        # residual-based selections during SO
6 }

```

Following Weng et al. [2020], a common algebraic operator set is used across all splits as given below.

SISSO++ operator set

```

1 # add: z1 + z2, sub: z1 - z2
2 # mult: z1 * z2, div: z1 / z2
3 # sqrt: sqrt(z)
4
5 sisso_operators = [
6     "add",
7     "sub",
8     "mult",
9     "div",
10    "sqrt",
11 ]

```

This operator library generates descriptors such as products, differences, and ratios of primary and recursively constructed features. In particular, it allows **SISSO++** to represent structural quantities closely related to the scientifically important octahedral-to-tolerance-factor ratio μ/t identified by Weng et al. [2020]. As **SISSO++** constructs its candidate descriptor space through complete enumeration, we adopt moderate search settings to prevent the rapid combinatorial growth of symbolic features and avoid excessive computational overhead.

After **SIS** and **S0** steps, the final recovered expression is formed as a sparse linear combination of the selected symbolic descriptors returned by **SISSORegressor**.

B.5 gplearn

We use the `gplearn.genetic.SymbolicRegressor` (<https://gplearn.readthedocs.io/>) implementation of **gplearn** [Stephens, 2016]. Following the original study in Weng et al. [2020], we consider the same set of structural and electronic descriptors $\{R_A, \chi_A, \chi_B, Q_A, N_d, \mu, t, \rho\}$ as candidate variables, additionally with the current density (ρ) levels. We consider 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, **gplearn** is fitted to the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $\text{MAE} = n^{-1} \sum_{i=1}^n |y_i - \hat{y}_i|$.

Following Weng et al. [2020], the operator set used across all splits is configured as follows.

gplearn operator set

```

1 function_set = [
2     "add",    # (x, y) -> x + y
3     "sub",    # (x, y) -> x - y
4     "mul",    # (x, y) -> x * y
5     "div",    # protected division
6     "sqrt",   # protected square root
7 ]

```

The protected implementations of `div` and `sqrt` are adopted to avoid numerical instabilities during symbolic evolution, following the default **gplearn** implementation. We maintain similar

gplearn settings as adopted in Weng et al. [2020] across all 5 splits of the dataset.

gplearn settings

```

1 # The settings closely follow those adopted in
2 # Weng et al. (2020, Nature Communications)
3
4 gplearn_settings = {
5     "population_size": 5000,          # symbolic programs per generation
6     "generations": 20,                # maximum number of generations
7     "stopping_criteria": 0.01,        # early stopping threshold for MAE
8     "p_crossover": 0.5,               # subtree crossover probability
9     "p_subtree_mutation": 1 / 6,      # subtree mutation probability
10    "p_hoist_mutation": 1 / 6,        # hoist mutation probability
11    "p_point_mutation": 1 / 6,        # point mutation probability
12    "max_samples": 0.95,               # fraction of training samples per program
13    "tournament_size": 20,             # tournament selection size
14    "parsimony_coefficient": 5e-4,     # complexity penalty
15    "metric": "mean absolute error",   # fitness criterion, MAE
16    "constant_range": (-1, 1),        # range of ephemeral constants
17 }
```

The final recovered symbolic expression is extracted from the fitted estimator through

Extracting the final gplearn expression

```
1 model._program
```

which stores the best symbolic program obtained after the final generation.

B.6 operon

We use the `pyoperon.sklearn.SymbolicRegressor` implementation (<https://github.com/heal-research/operon>) of `operon` [Burlacu et al., 2020]. We consider 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, `operon` is fitted to the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $MAE = n^{-1} \sum_{i=1}^n |V_{RHE,i} - \hat{V}_{RHE,i}|$.

Following Weng et al. [2020], the primitive symbol set across all splits is configured as follows.

operon operator set

```

1 allowed_symbols = [
2     "add",      # (x, y) -> x + y
3     "sub",      # (x, y) -> x - y
4     "mul",      # (x, y) -> x*y
5     "div",      # (x, y) -> x/y
6     "sqrt",     # x -> sqrt(x)
7     "constant", # numerical-constant leaf nodes
8     "variable", # input-variable leaf nodes
```

continued on next page

continued from previous page

9]

For all 5 train/test splits, **operon** is run with the following common configuration.

operon settings

```

1 # generations: number of evolutionary generations;
2 # population_size: number of candidate expressions maintained in the population;
3 # max_length: maximum length of the evolved expressions
4
5 operon_settings = {
6     "generations": 2000,
7     "population_size": 1000,
8     "max_length": 20,
9 }
```

The estimator is initialized separately for each split.

operon module initialization

```

1 model = SymbolicRegressor(
2     allowed_symbols=allowed_symbols,
3     generations=2000,
4     population_size=1000,
5     max_length=20,
6 )
```

The final recovered expression is extracted from the fitted estimator through

Extracting the final operon expression

```

1 model.get_model_string(model.model_)
```

when the fitted expression is stored in `model_`; otherwise, the available model string attribute returned by the fitted estimator is used.

B.7 PySR

We use the `pysr.PySRRegressor` implementation (<https://github.com/MilesCranmer/PySR>) of PySR [Cranmer, 2023]. We consider 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, PySR is fitted to the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $\text{MAE} = n^{-1} \sum_{i=1}^n |V_{\text{RHE},i} - \hat{V}_{\text{RHE},i}|$.

Following [Weng et al., 2020], the operator set used across all splits is configured as follows.

PySR operator set

```

1 binary_operators = [
2     "+",      # (x, y) -> x + y
3     "-",      # (x, y) -> x - y
4     "*",      # (x, y) -> x*y
5     "/",      # (x, y) -> x/y
6 ]
7
8 unary_operators = [
9     "sqrt",    # x -> sqrt(x)
10 ]

```

For all 5 splits of the dataset, PySR is run with the following common settings.

PySR settings

```

1 # niterations: number of evolutionary search iterations;
2 # populations: number of populations evolved during the search;
3 # population_size: number of candidate expressions in each population;
4 # maxsize: maximum permitted expression tree size;
5 # elementwise_loss: loss used to fit and rank candidate expressions;
6 # model_selection: rule used to select the final expression.
7
8 pysr_settings = {
9     "niterations": 500,
10    "populations": 31,      # PySR default
11    "population_size": 27, # PySR default
12    "maxsize": 20,
13    "elementwise_loss": "loss(x, y) = (x - y)^2", # squared-error loss
14    "model_selection": "best",
15    "deterministic": True,
16    "parallelism": "serial",
17 }

```

The final recovered expression is first extracted through the SymPy [Meurer et al., 2017] interface,

Extracting the final PySR expression

```

1 model.sympy()

```

and, if symbolic conversion is unavailable, the equation of the best-ranked model is obtained as follows.

Alternative extraction of the best PySR model

```

1 model.get_best()["equation"]

```

B.8 BMS

We use the `BMSRegressor` implementation from the `AutoRA` module (<https://autoresearch.github.io/autora/user-guide/theorists/bms/>) [Musslick et al., 2024] of Bayesian machine scientist (BMS) [Guimerà et al., 2020]. We consider 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, BMS is fitted on the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $\text{MAE} = n^{-1} \sum_{i=1}^n |V_{\text{RHE},i} - \hat{V}_{\text{RHE},i}|$.

We run 5 parallel MCMC chains of BMS for each train/test split of the dataset. Following Weng et al. [2020], across all splits, the BMS operator set is configured using the primitive operators supported by the `BMSRegressor` interface.

BMS operator set

```
1 bms_operator_set = [
2     "add",      # (x, y) -> x + y
3     "sub",      # (x, y) -> x - y
4     "mul",      # (x, y) -> x*y
5     "div",      # (x, y) -> x/y
6     "psqrt",    # x -> sqrt(abs(x))
7 ]
```

The aforementioned operators are passed through the `ops` dictionary, with a uniform prior weight assigned to each included operator.

BMS operator dictionary and prior

```
1 bms_ops = {
2     "+": 2,
3     "-": 1,
4     "*": 2,
5     "/": 2,
6     "psqrt": 1,
7 }
8
9 bms_prior = {
10    "Nopi_+": 1.0,
11    "Nopi_-": 1.0,
12    "Nopi_*": 1.0,
13    "Nopi_/": 1.0,
14    "Nopi_psqrt": 1.0,
15 }
```

For each train/test split, the 5 MCMC parallel chains are comprise `epochs = 2000` MCMC epochs.

BMS settings

```
1 bms_settings = {
2     "n_parallel_chains": 5, # independent MCMC chains per split
3     "epochs": 2000,        # MCMC steps within each chain
```

continued on next page

continued from previous page

```

4     "parallel_chains": True,
5 }

```

After all 5 parallel chains are completed, we retain the chain attaining the lowest in-sample RMSE on the corresponding 90% training set. Its symbolic expression is reported as the final recovered BMS model.

Extracting the final BMS expression across parallel chains

```

1 best_chain = min(
2     successful_chains,
3     key=lambda chain: chain["chain_train_rmse"],
4 )
5
6 final_expression = best_chain["expression"]

```

B.9 BSR

We use the BSRRegressor implementation from the AutoRA module (<https://autoresearch.github.io/autora/user-guide/theorists/bsr/>) [Musslick et al., 2024] of Bayesian symbolic regression (BSR) [Jin et al., 2020]. We consider 5 independent 90/10 train/test splits of the oxide perovskite dataset. For each split, BSR is fitted on the corresponding 90% training set, while predictive performance is evaluated using train and held-out 10% test RMSE and MAE, respectively. MAE denotes mean absolute error, i.e., $\text{MAE} = n^{-1} \sum_{i=1}^n |V_{\text{RHE},i} - \hat{V}_{\text{RHE},i}|$.

We run 5 independent BSR restarts for each train/test split of the dataset. Following Weng et al. [2020], across all splits, the BSR operator set is configured as follows.

BSR operator set

```

1 bsr_operator_set = [
2     "add",      # (x, y) -> x + y
3     "sub",      # (x, y) -> x - y
4     "mul",      # (x, y) -> x*y
5     "inv",      # x -> 1/x
6     "psqrt",    # x -> sqrt(abs(x))
7 ]

```

The aforementioned operator set is converted into the following prior dictionary, with equal prior weights assigned to all included operators.

BSR operator prior dictionary

```

1 bsr_operator_prior = {
2     "+": 1.0,
3     "-": 1.0,
4     "*": 1.0,
5     "inv": 1.0,

```

continued on next page

continued from previous page

```

6     "psqrt": 1.0,
7 }

```

For each train/test split, the 5 restarts employ $K = 8$ symbolic trees similar to the **HIERBOSSS** configuration described in [Section B.1](#). The split probability parameters are maintained at their default specifications.

BSR settings

```

1 # split-probability parameters maintained at default settings
2
3 bsr_settings = {
4     "n_restarts": 5,                # independent fits per split
5     "itr_num": 2000,               # iterations within each restart
6     "tree_num": 8,                # additive symbolic trees
7     "prior_name": "perovskites_common_ops",
8 }

```

After completing all 5 restarts, we retain the run attaining the lowest in-sample RMSE on the corresponding 90% training set. The final recovered model is the additive symbolic expression returned by that restart, together with the fitted coefficients of its selected symbolic trees.

Extracting the final BSR expression across restarts

```

1 best_restart = min(
2     successful_restarts,
3     key=lambda restart: restart["chain_train_rmse"],
4 )
5 final_expression = best_restart["expression"]

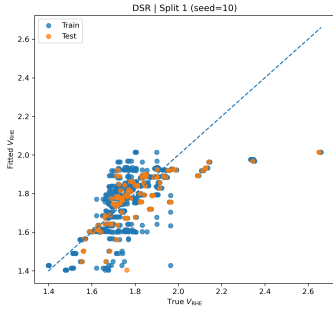
```

C Results of Competitors for the Oxide Perovskite Data Study

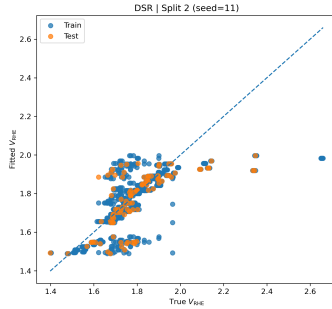
C.1 Results of DSR

Table 2: DSR results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity.

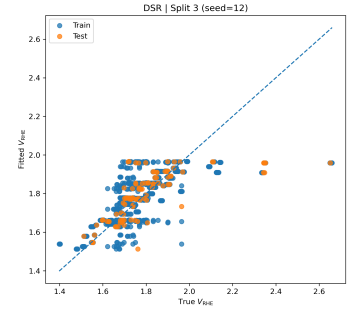
Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$\sqrt{\mu + \sqrt{\frac{Q_A(Q_A + \sqrt{p})}{\chi_A \chi_B}}}$	0.117	0.133	0.077	0.082	32.031
2	$\sqrt{\mu \left[\sqrt{R_A + p} + \mu \sqrt{R_A + \sqrt{\mu} + R_A} \right] + \chi_A}$	0.122	0.128	0.075	0.087	31.175
3	$\sqrt{\sqrt{t + \frac{t Q_A^2 (1 + p + Q_A + \chi_B + t)}{\chi_A}}}$	0.116	0.139	0.075	0.082	33.378
4	$\sqrt{R_A} \frac{\mu}{\sqrt{\mu}} \left[\sqrt{\sqrt{\sqrt{t} - p} + \chi_A} \right]$	0.120	0.142	0.077	0.079	35.367
5	$\sqrt{\mu + \frac{\sqrt{\sqrt{Q_A} + Q_A + 2R_A + \chi_A + p} + \sqrt{R_A}}{\chi_B}}$	0.114	0.128	0.081	0.086	31.566



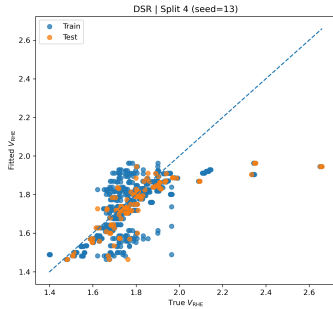
(a) Split 1



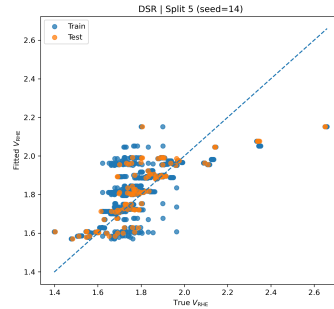
(b) Split 2



(c) Split 3



(d) Split 4



(e) Split 5

Figure 2: True versus fitted V_{RHE} values obtained by DSR across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

C.2 Results of QLattice

Table 3: QLattice results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity.

Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$0.669 (0.074 \rho + 0.359) \left(0.066 N_d - 1.018 + \frac{1}{1.196 \chi_B - 1.121} \right) + 1.569$	0.095	0.084	0.066	0.061	84.321
2	$0.030 \rho - 0.360 (1.365 \chi_B - 2.858) (7.387 \chi_B - 11.859) \times (-0.076 \rho + 16.542 \mu - 6.813) + 1.659$	0.094	0.092	0.062	0.067	79.871
3	$-1.139 t + 0.096 (17.657 - 8.738 \chi_B) (0.073 \rho - 0.031) \times [1.051 \chi_B + (0.385 N_d - 2.099)] \times (0.823 N_d - 3.891) - 0.754 + 2.811$	0.093	0.090	0.062	0.057	85.091
4	$3.233 \mu - 0.078 (0.180 \rho - 1.021) \times [0.122 N_d + (2.519 - 0.328 N_d) \times (2.080 \chi_B - 3.561) (0.119 \rho + 0.938) - 2.143] + 0.416$	0.091	0.101	0.061	0.061	84.629
5	$0.011 \rho + 3.347 \mu - 0.083 (1.095 - 0.289 \rho) (19.800 - 18.681 t) \times (22.870 - 12.463 \chi_B) (23.077 - 23.015 t) + 0.302$	0.089	0.106	0.058	0.070	90.842

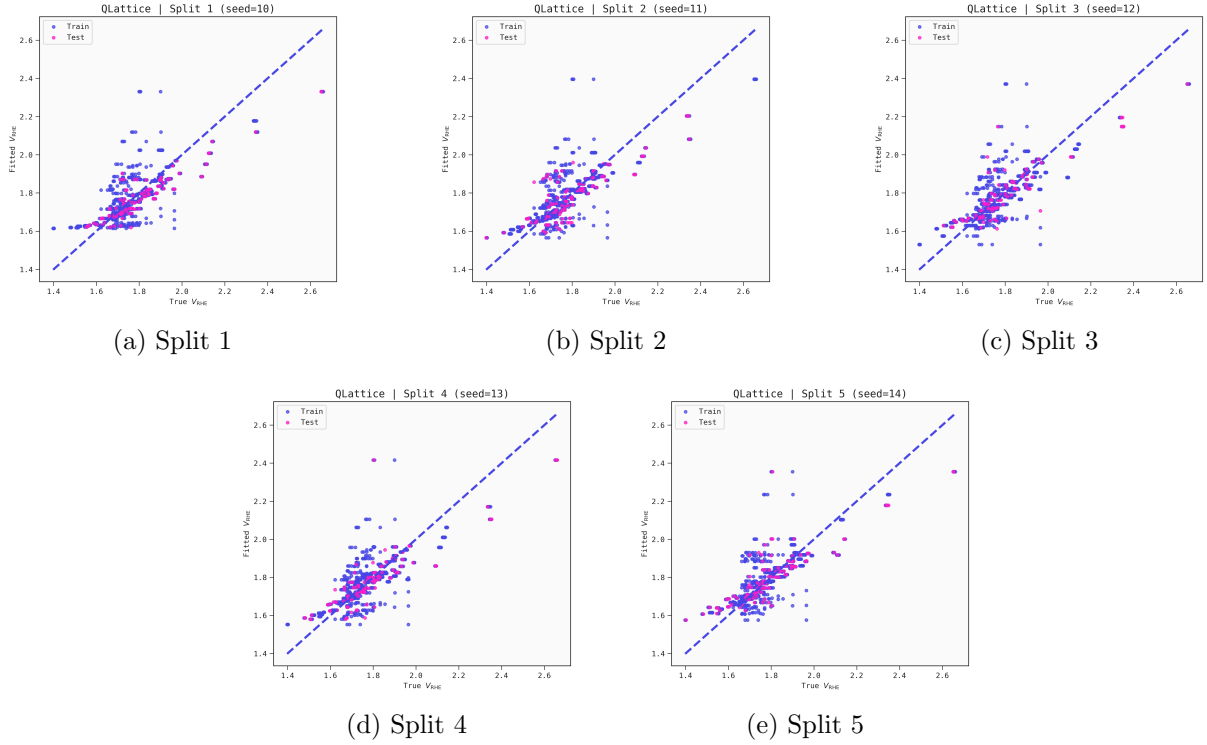
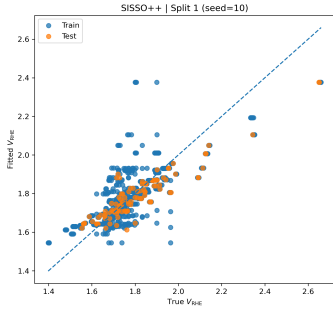


Figure 3: True versus fitted V_{RHE} values obtained by QLattice across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

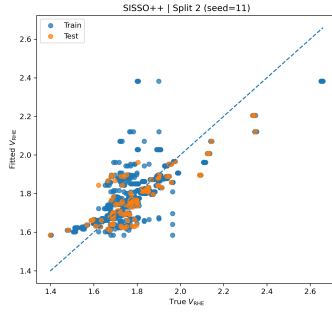
C.3 Results of SISSO++

Table 4: SISSO++ results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity.

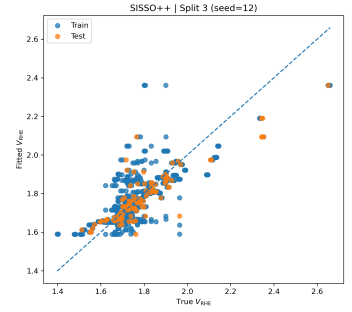
Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$1.237 - 0.075 \left[\rho t - (\rho - \chi_A) \right]$ $+ 1.507 \left(\frac{\mu}{t} - \frac{\mu}{N_d} \right) + 0.006 \frac{\rho \chi_A}{\chi_B - R_A}$	0.094	0.083	0.064	0.058	3.971
2	$1.429 + 0.003(\rho + N_d)(N_d - Q_A)$ $+ 0.067 \frac{\rho + \chi_B}{\chi_B - \chi_A} - 0.034 \frac{\rho \chi_B}{\chi_B - \chi_A}$	0.094	0.090	0.062	0.067	4.989
3	$1.534 + 0.205 \frac{\rho/N_d}{\chi_B - \chi_A}$ $+ 0.025(\rho + Q_A) \frac{Q_A}{\chi_B} - 0.266 \frac{\rho Q_A}{\chi_B N_d}$	0.093	0.095	0.063	0.060	6.363
4	$1.070 - 0.088 \left[\rho t - (\rho - \chi_A) \right]$ $+ 2.061 \frac{\mu N_d}{N_d + R_A} + 0.006 \frac{\rho \chi_A}{\chi_B - R_A}$	0.092	0.099	0.063	0.061	5.706
5	$1.546 + 0.001 \rho N_d (N_d - Q_A)$ $+ 0.067 \frac{\rho + \chi_A}{\chi_B - \chi_A} - 0.034 \frac{\rho \chi_B}{\chi_B - \chi_A}$	0.091	0.108	0.062	0.072	5.764



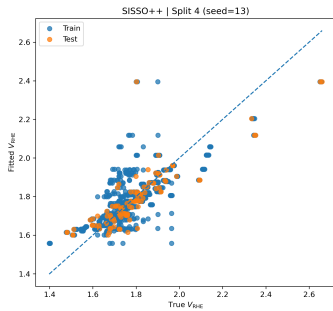
(a) Split 1



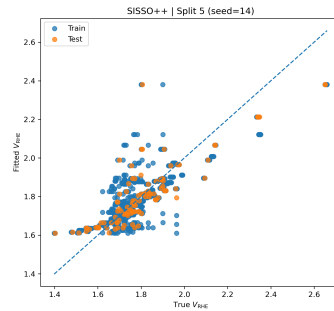
(b) Split 2



(c) Split 3



(d) Split 4



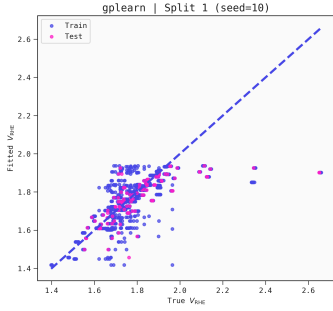
(e) Split 5

Figure 4: True versus fitted V_{RHE} values obtained by SISSO++ across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

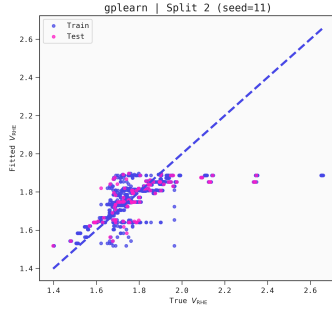
C.4 Results of gplearn

Table 5: **gplearn** results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Descriptors highlighted in — were identified by [Weng et al. \[2020\]](#)[Table 2] as particularly important for OER activity.

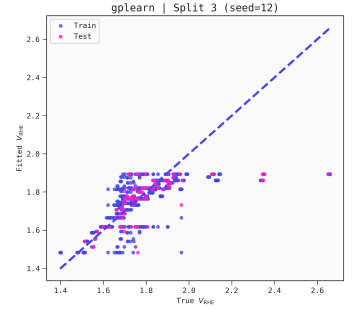
Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$\sqrt{\sqrt{Q_A} \sqrt{\sqrt{N_d + \frac{p}{\chi_A^3}}}}$	0.124	0.140	0.073	0.074	43.128
2	$\sqrt{\sqrt{\sqrt{Q_A + p} + \frac{\chi_A}{0.803}}}$	0.124	0.125	0.072	0.082	41.212
3	$\sqrt{\sqrt{\sqrt{2p + 2Q_A + 0.523}}}$	0.122	0.150	0.072	0.081	42.081
4	$\sqrt{\sqrt{\sqrt{\chi_B - Q_A - \frac{p}{R_A} + t + \mu}}}$	0.118	0.143	0.070	0.075	43.619
5	$\sqrt{\sqrt{\sqrt{\frac{p}{t} + 2Q_A - R_A + \mu}}}$	0.115	0.154	0.066	0.083	41.193



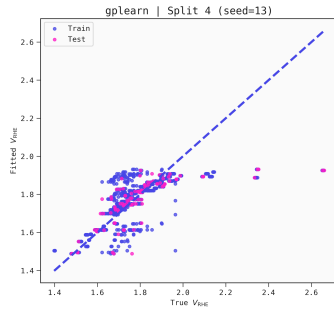
(a) Split 1



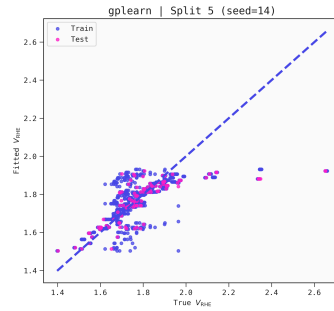
(b) Split 2



(c) Split 3



(d) Split 4



(e) Split 5

Figure 5: True versus fitted V_{RHE} values obtained by **gplearn** across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

C.5 Results of operon

Table 6: **operon** results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity.

Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$1.736 + 0.007 \left[\frac{1.670 Q_A}{0.382 \mu - 1.097 \chi_B} + \frac{\mathcal{A}_1}{1.772 \chi_B} \right],$ $\mathcal{A}_1 = \frac{1.134 \chi_B}{1.687 \mu + 1.921} - \frac{0.200 t Q_A}{0.007}$ $+ \frac{1.213 \mu N_d}{1.221 - 1.097 \chi_B} + \frac{\mathcal{B}_1}{\sqrt{1.558} - 1.097 \chi_B},$ $\mathcal{B}_1 = (0.382 \mu \chi_A)^2 (0.790 Q_A + 1.097 \chi_B) (0.166 \chi_A) + 1.389 \mu$	0.090	0.082	0.060	0.057	27.062
2	$1.485 + 0.001 \mathcal{A}_2,$ $\mathcal{A}_2 = -14.338 t N_d + 7.516 Q_A + 1.107 \mu$ $+ 8.138 Q_A N_d - \frac{\mathcal{B}_2}{C_2} + \sqrt{4.453 \mu} (0.923 N_d)$ $+ \frac{2.718 N_d + 2.223 \mu}{(0.274 \chi_B)^4 (0.923 \chi_B)},$ $\mathcal{B}_2 = \frac{1.414 Q_A}{(0.274 R_A)^3},$ $C_2 = \frac{-1.155 \mu (2.718 Q_A + 0.854 \mu)}{0.923 \chi_B} + 1.618 N_d$	0.090	0.088	0.058	0.063	29.585
3	$1.459 + 0.000 \mathcal{A}_3,$ $\mathcal{A}_3 = \frac{\frac{1.446 \mu + 3.142 Q_A}{0.102 / (1.952 \mu) / \sqrt{0.536 \mu}}}{\frac{0.176 \chi_B}{4.008} - \frac{1}{-0.070 \chi_B}}$ $+ \frac{\sqrt{0.498 \mu}}{0.571 \chi_B (-0.131 \chi_B)} + \frac{1.367 \mu}{0.975}$ $+ \frac{(0.497 \chi_B)(0.542 \chi_B)(-0.070 \chi_B)}{0.102 / (1.501 \chi_B)} + \frac{3.337 N_d}{3.764 Q_A}$ $+ \frac{-0.063 \mu + \frac{(-0.838 \chi_B) / (-0.084 \chi_B)}{0.651 \mu}}{0.651 \mu}$	0.091	0.095	0.062	0.062	28.378
4	$1.947 + 0.253 (\mathcal{A}_4 - \mathcal{B}_4),$ $\mathcal{A}_4 = \frac{(0.219 Q_A + 0.179 \mu)(0.257 N_d)}{(0.147 \chi_B - 0.345 \mu)(0.318 \chi_B)(5.830 \chi_B)(0.434 \chi_B)},$ $\mathcal{B}_4 = \frac{-0.909 \mu / 0.743}{-1.065 + 0.179 \mu - 2.692 t} - \frac{0.693 Q_A}{3.138 \chi_B}$ $+ \frac{-0.910 \mu^2}{(1.901 \mu + 0.179 \mu - 2.692 t)(3.138 \chi_B)} - \frac{10.141}{3.138 \chi_B}$	0.089	0.099	0.061	0.065	26.507
5	$1.639 + 0.001 \mathcal{A}_5,$ $\mathcal{A}_5 = \frac{-0.579 \chi_B + 0.602 N_d}{-0.435 + 0.853 \mu}$ $- \frac{(0.825 - 0.175 Q_A)}{(-0.435)(-0.459)(-0.435 + 0.853 \mu)}$ $- \frac{1.187 \mu}{(4.182 \chi_B)(1.293 \mu)(-0.435 + 0.872 \mu)(-0.421 \chi_B)}$ $- \frac{0.602 t - 0.175 Q_A}{1.247 \mu} - \frac{1}{4.970 \chi_B / \frac{0.602 t - 0.175 Q_A}{-0.298 R_A} - 0.579 \chi_B + 1.293 \mu}$	0.090	0.107	0.061	0.073	26.964

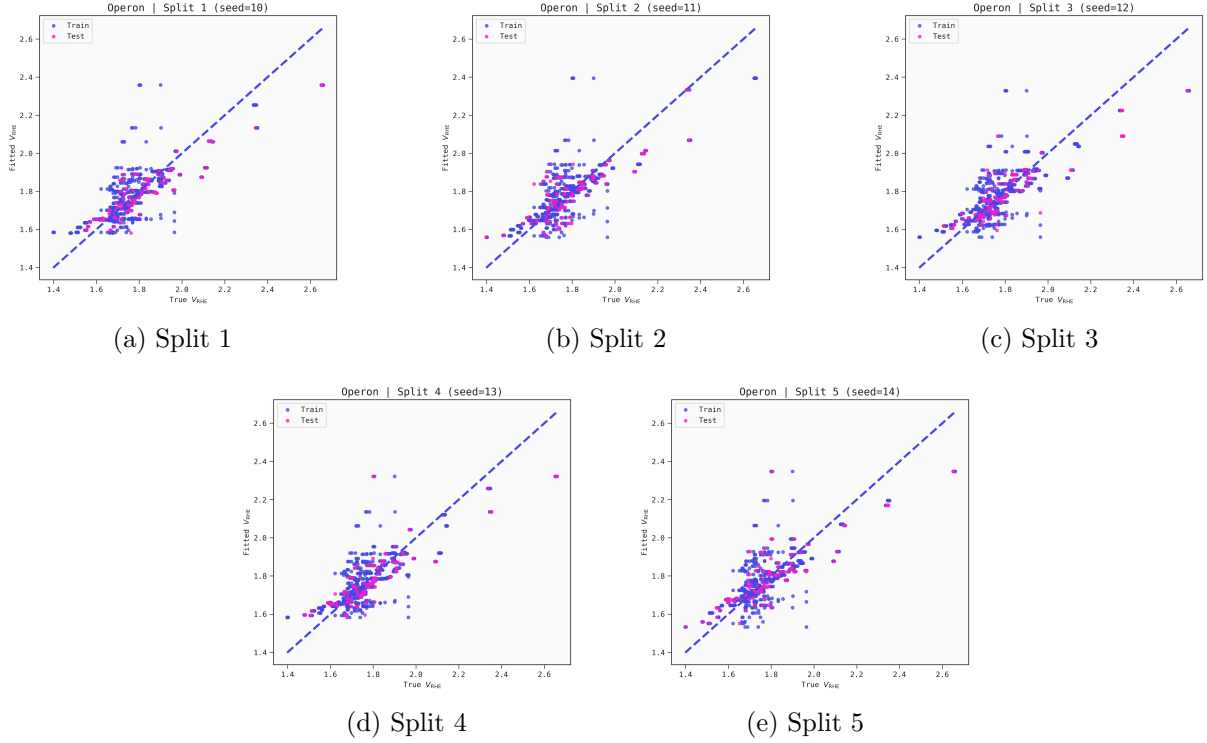


Figure 6: True versus fitted V_{RHE} values obtained by **operon** across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

C.6 Results of PySR

Table 7: PySR results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity.

Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$0.015 \rho + 4.011 \mu$	0.107	0.111	0.066	0.060	156.604
2	$0.016 \rho + 3.998 \mu$	0.108	0.109	0.065	0.075	157.797
3	$0.016 \rho + 4.001 \mu$	0.106	0.121	0.066	0.068	158.251
4	$0.016 \rho + 4.001 \mu$	0.106	0.119	0.066	0.066	157.644
5	$0.016 \rho + 4.005 \mu$	0.104	0.137	0.064	0.083	157.306

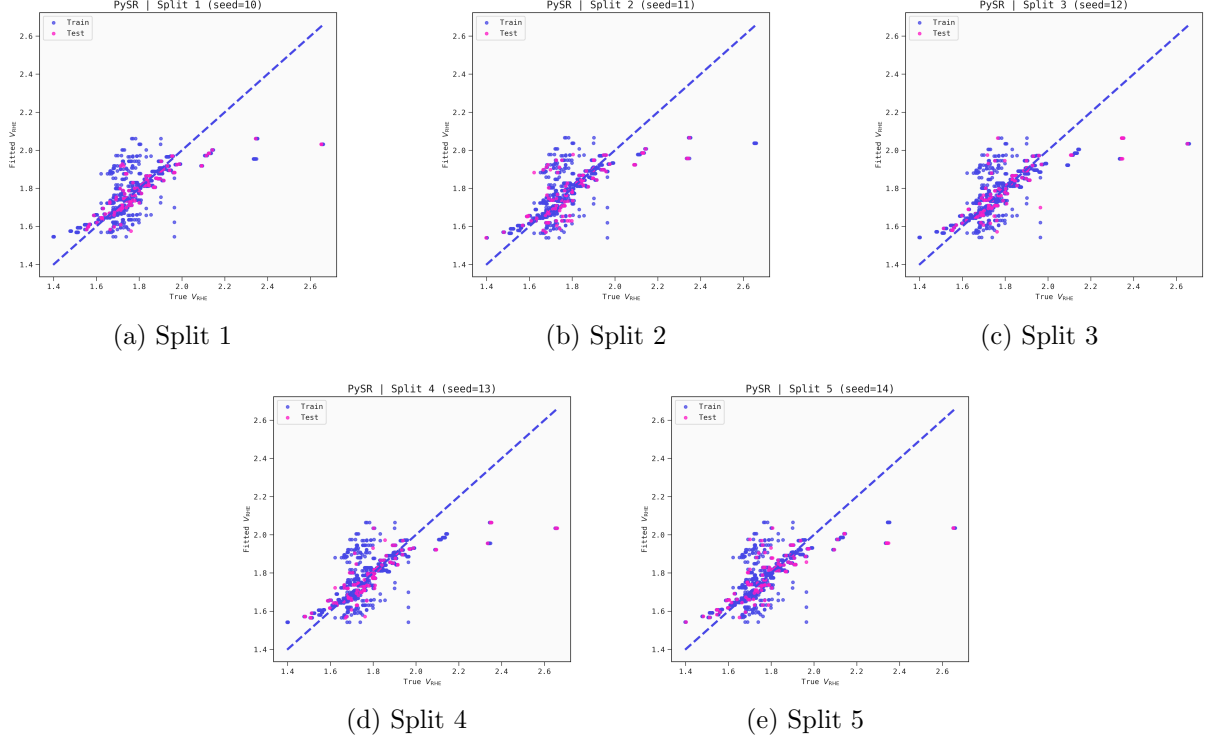


Figure 7: True versus fitted V_{RHE} values obtained by PySR across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

C.7 Results of BMS

Table 8: BMS results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Here, a_0 denotes the BMS fitted constant. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity.

Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$\mu \sqrt{\left a_0 + \mu \left[\frac{\mu}{\rho (\sqrt{ \mathcal{A}_1 } + \mathcal{B}_1)} + \frac{\rho}{\chi_A \sqrt{ \chi_B }} \right] \right },$ $\mathcal{A}_1 = \frac{t}{N_d(-\rho)} \chi_B,$ $\mathcal{B}_1 = (\rho - R_A N_d) \frac{(Q_A + \chi_A)/(-\rho)}{\rho + \sqrt{ \chi_B } + \mu}$	0.095	0.077	0.064	0.056	352.474
2	$-a_0 \mu \sqrt{\left \sqrt{\sqrt{ \mathcal{A}_2 }} \right },$ $\mathcal{A}_2 = \frac{R_A \sqrt{a_0} \chi_B}{\mu} + \frac{(\rho + a_0) \mathcal{B}_2}{(\chi_B - a_0 \chi_A) R_A} + \rho,$ $\mathcal{B}_2 = \frac{\rho}{N_d(\chi_B + \mu)} \frac{1}{(\chi_B + a_0) R_A + 2t}$	0.094	0.092	0.062	0.068	429.338

Continued on next page

Table 8: BMS results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Here, a_0 denotes the BMS fitted constant. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity (*continued*).

Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
3	$\frac{\sqrt{ -\mu }}{\sqrt{(a_0 + R_A) [-\chi_B + \sqrt{ \mathcal{A}_3 }]}} + \mu + \sqrt{ a_0 },$ $\mathcal{A}_3 = \frac{\sqrt{\left \sqrt{ \rho } + \left(\rho + \frac{\chi_B}{R_A/N_d} \right) \right }}{\mathcal{B}_3},$ $\mathcal{B}_3 = \frac{t/\mu}{\rho + a_0} + t + \frac{1}{\mu\rho + \mu}$	0.095	0.096	0.066	0.065	504.364
4	$-\mu \left[-\sqrt{ a_0 } + \frac{\rho/\mathcal{A}_4}{-N_d} + \frac{\mu}{a_0} \right],$ $\mathcal{A}_4 = \frac{\sqrt{\left \sqrt{ \mathcal{B}_4 + \sqrt{a_0\rho\chi_B} } \right } \sqrt{ \chi_B }}{\chi_A/\chi_B},$ $\mathcal{B}_4 = \sqrt{(-\chi_A) \frac{a_0}{\rho} \sqrt{ R_A }} \frac{\chi_A + Q_A}{R_A + t}$	0.098	0.104	0.071	0.067	415.098
5	$\sqrt{\left \frac{\sqrt{ Q_A/a_0 }}{-\chi_B + \mathcal{A}_5} \right } + \sqrt{ \mu },$ $\mathcal{A}_5 = \frac{\sqrt{\left \rho + \chi_B \sqrt{ \rho } \right }}{\chi_B(a_0 + R_A)} a_0$	0.095	0.108	0.066	0.074	391.923

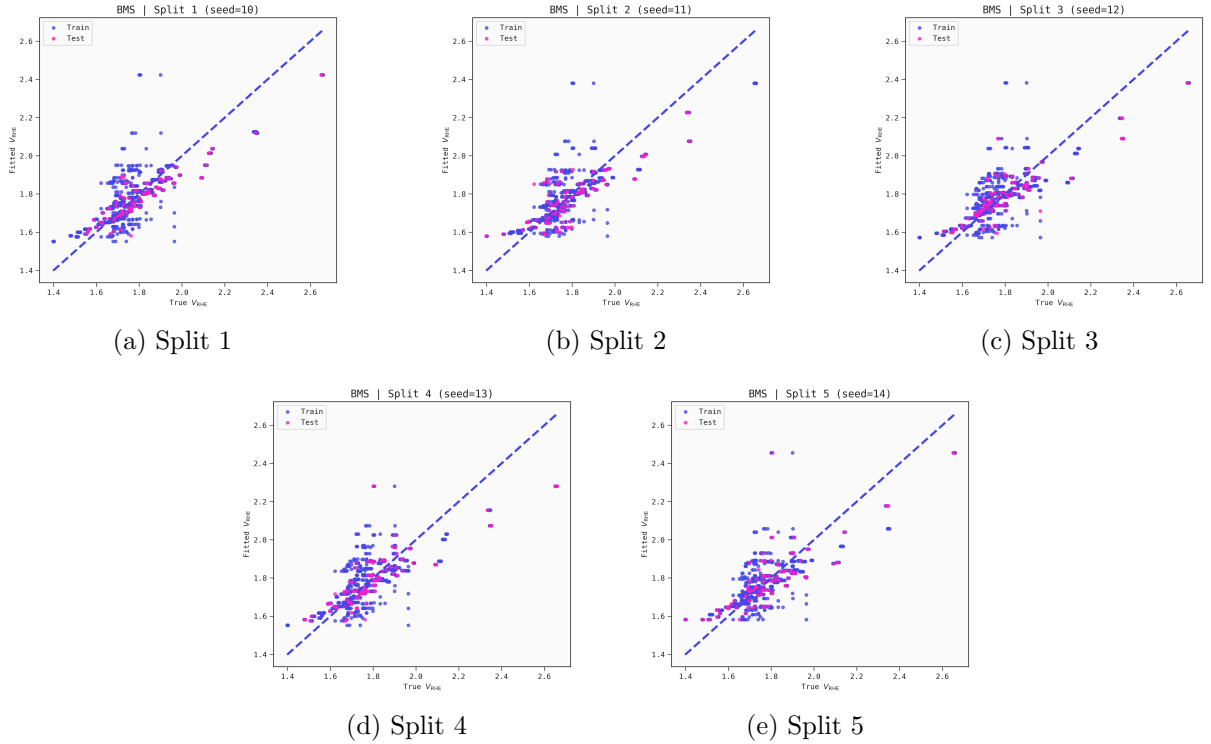


Figure 8: True versus fitted V_{RHE} values obtained by BMS across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

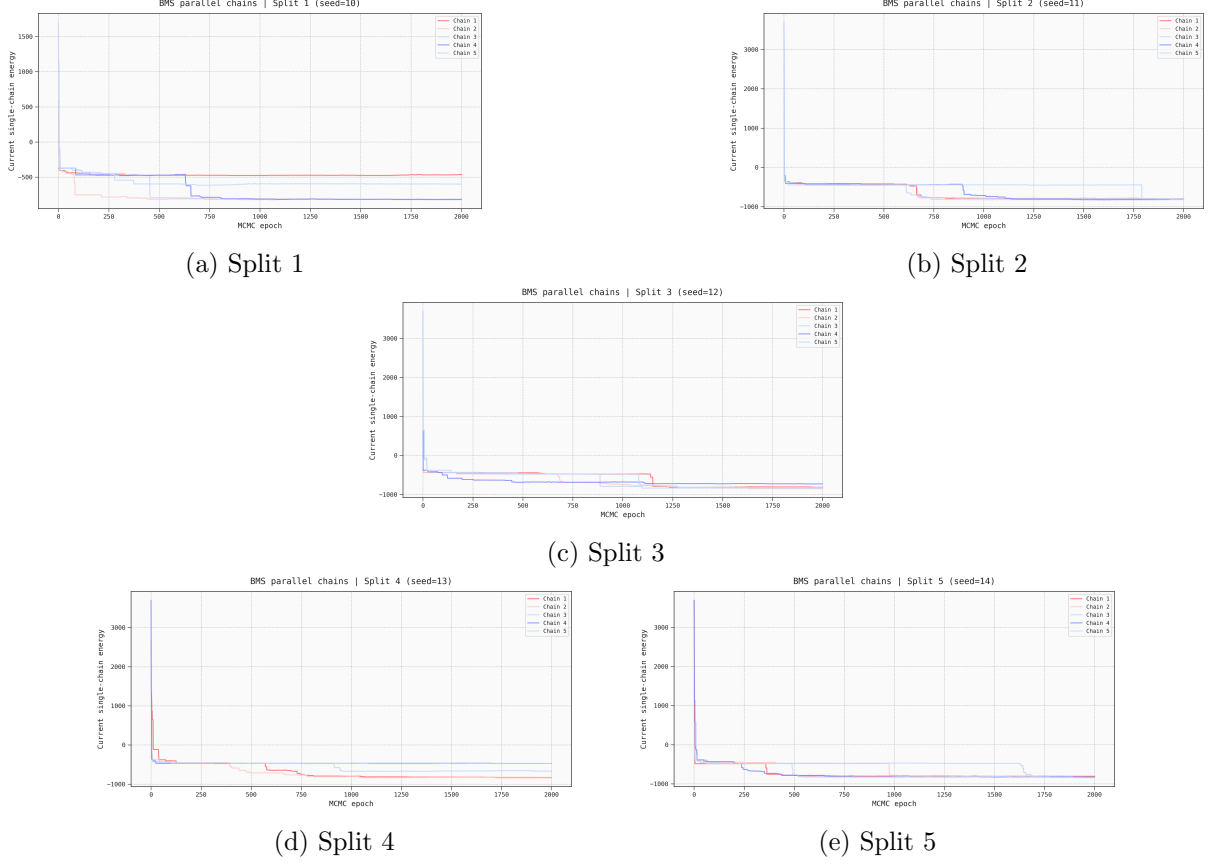


Figure 9: Trace plots of BMS across the 5 splits of the oxide perovskite dataset.

C.8 Results of BSR

Table 9: BSR results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity.

Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$-0.684 - 0.497(\rho - Q_A) + 1.713R_A N_d$					
	$-0.708 \left(\chi_B + \frac{1}{N_d} \right) - 0.496 \left\{ \frac{1}{1/(\rho - Q_A)} \right\}$					
	$-0.049(Q_A + \mu - \mu) + \frac{1.563}{\chi_A}$	0.103	0.104	0.069	0.062	525.408
	$-0.714 (Q_A + R_A^2 N_d - \mu)$					
	$+1.008(\rho - N_d - R_A)$					
2	$-1.913 + 0.023 \rho \mu + 2.450 \left(\frac{1}{\chi_A} + \chi_A \right)$					
	$-0.163 (\mu N_d + \chi_B + R_A)$					
	$-0.627 \left(\frac{1}{R_A + N_d} + \chi_A \right)$	0.104	0.102	0.068	0.072	510.201
	$-0.001 \left[\frac{1}{\rho} + \chi_A + \frac{1}{R_A} \left(t - \frac{1}{t} + \chi_B \right) \right]$					
	$+0.281(N_d - \mu) - 0.109 t^2 \chi_B N_d$					
	$+0.002 Q_A (\mu + \rho)$					

Continued on next page

Table 9: BSR results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. Descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity (*continued*).

Split	Learned expression	Train RMSE	Test RMSE	Train MAE	Test MAE	Runtime (s)
3	$1.748 + 0.132 Q_A \mu$ $+ \frac{0.028}{\rho t - \chi_B} + 0.003(\rho + Q_A)N_d$ $- \frac{0.042}{R_A} - \frac{0.025}{(1/t)R_A}$ $+ \frac{0.0002}{1/t - \chi_A}$ $- 0.241(R_A \chi_B + \mu - \chi_A) + \frac{1.179}{N_d}$	0.107	0.124	0.071	0.074	509.069
4	$1.788 \times 10^{-20} + 1.430 \times 10^{-11}(2\rho)$ $+ 7.232 \times 10^{-15} \mu(\mu + R_A - R_A)$ $- 2.356 \times 10^{-14}(\chi_B - R_A) - 7.117 \times 10^{-12}(\chi_A - \rho)$ $+ \frac{1.230 \times 10^{-14}}{R_A}$ $+ \frac{1.788 \times 10^{-10}}{(\chi_B - \chi_B)N_d}$ $- 1.593 \times 10^{-14}(t + t - t)$ $- \frac{2.263 \times 10^{-14}}{Q_A t}$	0.163	0.190	0.114	0.119	508.556
5	$1.231 - 0.001[\chi_A - (t - \rho + R_A)N_d]$ $- 0.0003 \chi_A \rho + 0.061 \frac{\rho}{Q_A}$ $+ 0.047 \left[Q_A \left(t + R_A + \frac{1}{\chi_B} \right) - R_A + t \right]$ $+ 0 \left[\frac{\chi_A - \chi_A}{t} \right] - \frac{0.824}{N_d + R_A}$ $+ \frac{1.429}{N_d t} + 0.170(\rho - t\rho)$	0.105	0.126	0.071	0.082	517.036

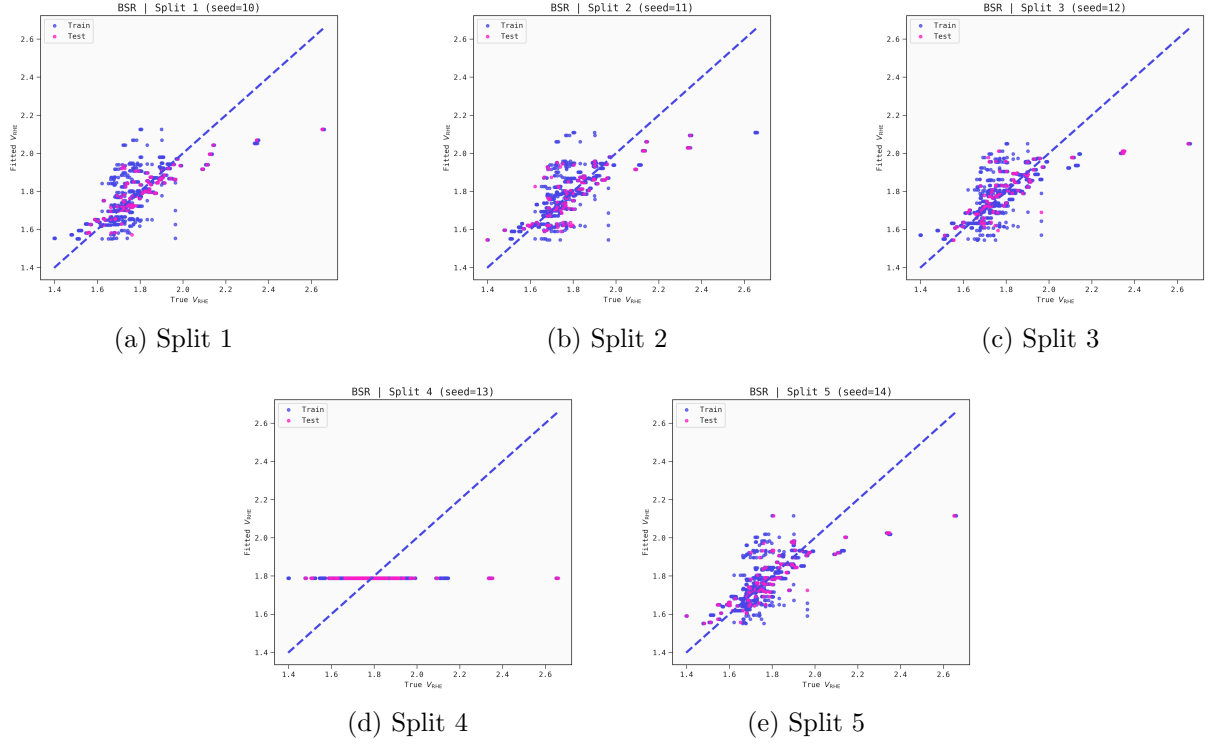


Figure 10: True versus fitted V_{RHE} values obtained by BSR across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

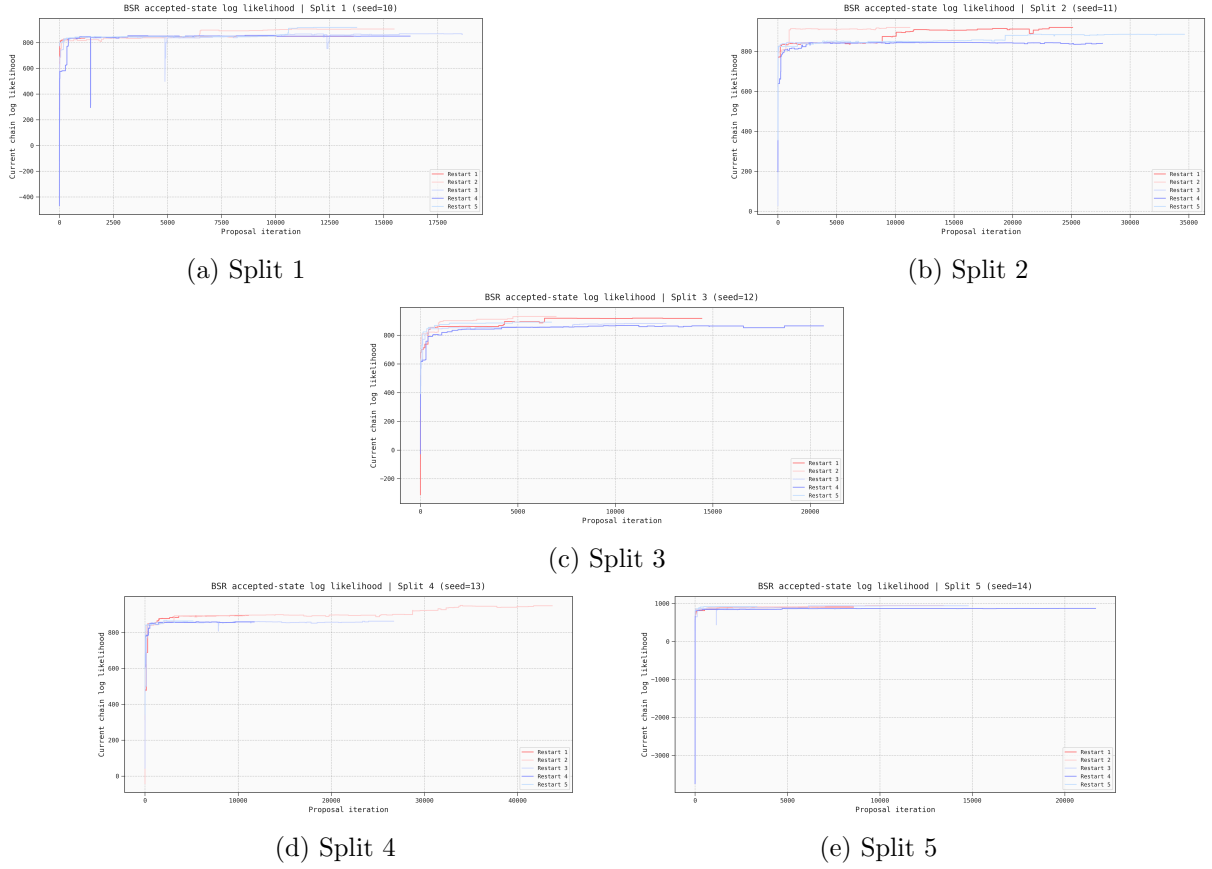


Figure 11: Trace plots of BSR across the 5 splits of the oxide perovskite dataset.

D Ablation Study for **HierBOSSS**: Practical Guidance for Choosing the Nominal Symbolic Forest Size

We conduct an ablation study: (i) to assess the sensitivity of **HierBOSSS** to the nominal number of symbolic trees K in the symbolic forest component $\mathcal{T}$ and (ii) as a practical guideline for selecting an appropriate nominal symbolic forest size that balances predictive performance with computational efficiency.

Following the experimental settings of **HierBOSSS** described in Section B.1, we consider symbolic forest sizes $K \in \{6, 7, 8, 10, 12\}$ and perform 5 independent 90/10 train/test splits of the oxide perovskite dataset. We report the predictive performance (mean test RMSE and MAE across 5 splits computed on the 10% held-out test set) of both the *Raw* and *Final* expressions of **HierBOSSS** together with the mean wall-clock runtime (over 5 splits) of the 5 parallel MCMC chains.

Figure 12 summarizes the predictive performance of **HierBOSSS** across different symbolic forest sizes. The average test RMSE and test MAE remain stable over the considered values of K , with only marginal improvements as the symbolic forest becomes larger. Furthermore, the predictive performance of the *Raw* and *Final* expressions is nearly indistinguishable, indicating that the post-MCMC symbolic model refinement step in **HierBOSSS** successfully removes redundant symbolic components while preserving predictive accuracy.

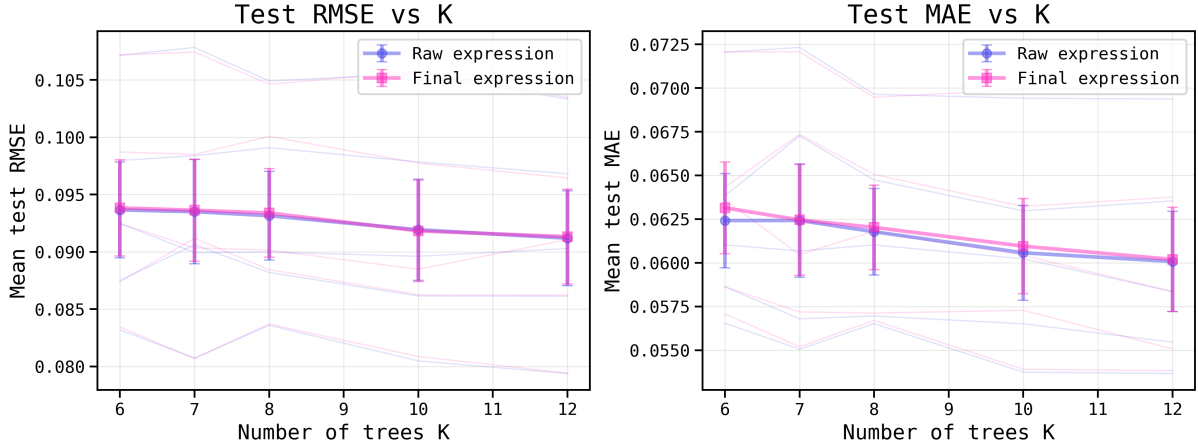


Figure 12: Average (over 5 splits) test RMSE and test MAE of **HierBOSSS** across different values of K for the oxide perovskite dataset.

The computational cost associated with increasing the symbolic forest size is shown in Figure 13. As expected, the mean parallel wall-clock runtime of running 5 parallel MCMC chains of **HierBOSSS** increases approximately monotonically with K , reflecting exploration of the larger symbolic search space. Although larger symbolic forests usually increase computational cost, they provide little additional predictive benefit once moderate values of K are reached.

Taken together, the preceding ablation results suggest a practical strategy for choosing the nominal symbolic forest size: select the smallest moderate value of K after which predictive accuracy stabilizes while avoiding unnecessarily large symbolic forests that incur additional computational cost. In the oxide perovskite data study, the predictive metrics stabilize around

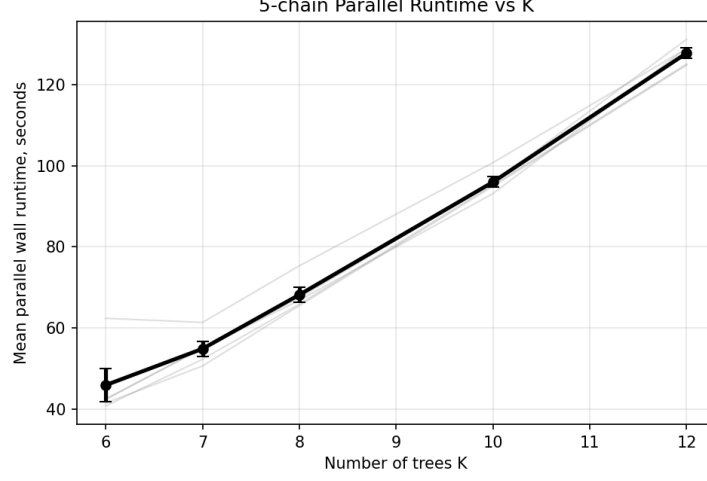


Figure 13: Runtime behavior of **HIERBOSS** across different values of K for the oxide perovskite dataset. The wall-clock runtime for 5 parallel chains across 5 splits has been reported, with the mean trend in black and split-level variability shown in gray.

$K \approx 8$, whereas the runtime continues to increase steadily for larger values of K . Consequently, a moderate symbolic forest provides sufficient representational flexibility for discovering informative symbolic descriptors, while the post-MCMC symbolic model refinement step in **HIERBOSS** data-adaptively removes redundant symbolic trees to produce a parsimonious final symbolic model and determines the effective forest size K_{eff} . Guided by this trade-off, we set $K = 8$ for the reported **HIERBOSS** analysis of the oxide perovskite dataset.

E Results of HierBOSSS for the Oxide Perovskite Data Study

E.1 Descriptors Learned by HierBOSSS

Table 10: HierBOSSS results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. *Raw* descriptor correspond to the symbolic forest learned by HierBOSSS followed by the *Final* descriptor after post-MCMC symbolic model refinement. The descriptors correspond to the top-ranked by JMP. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. *Final* descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity. Effective symbolic forest size denoted by K_{eff} .

Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
1	$\text{Raw: } 0.329 + 0.289\sqrt{\mu} + \frac{0.289}{\mu} - 0.004\rho^2$ $- \frac{0.674}{\sqrt{\chi_A} - \sqrt{\rho}} - 0.437\sqrt{\rho} + 0.006N_d\rho$ $+ \frac{0.310\rho}{\chi_B} + \frac{0.164\mu}{\rho}$ $\text{Final: } \frac{0.429}{\mu} - 0.004\rho^2 - \frac{0.811}{\sqrt{\chi_A} - \sqrt{\rho}}$ $- 0.454\sqrt{\rho} + 0.006N_d\rho + 0.339\rho\frac{1}{\chi_B} + \frac{0.206\mu}{\rho}$ $K_{\text{eff}}: 7$	0.090	0.084	0.060	0.057	89.277
2	$\text{Raw: } 0.297 + 0.236\sqrt{Q_A} + 0.357\sqrt{R_A} + 0.293\sqrt{\mu}$ $+ 0.335\sqrt{\chi_A} + \frac{0.538\rho}{N_d} + 0.097\sqrt{N_d}\rho$ $- 0.068\chi_B\rho - 0.076R_A\rho - 0.076\chi_A\rho$ $\text{Final: } 1.158 + 0.292\sqrt{Q_A} + \frac{0.530\rho}{N_d} + 0.095\sqrt{N_d}\rho$ $- 0.068\chi_B\rho - 0.073R_A\rho - 0.073\chi_A\rho$ $K_{\text{eff}}: 5$	0.092	0.086	0.060	0.064	66.506
3	$\text{Raw: } 0.219 + 0.357\mu + 0.352\sqrt{R_A} + 0.294\chi_A + \frac{0.338}{t}$ $+ 0.343\sqrt{\mu} + 0.006N_d\rho + \frac{0.290\rho}{\chi_B^2} - 0.257\mu\rho$ $\text{Final: } 2.103\sqrt{\mu} + 0.282\chi_A + 0.254\rho\frac{1}{\chi_B}\frac{1}{\chi_B}$ $+ 0.005N_d\rho - 0.214\mu\rho$ $K_{\text{eff}}: 5$	0.093	0.093	0.062	0.059	69.089
4	$\text{Raw: } 0.299 + 0.401\sqrt{Q_A + \rho} + \frac{0.356}{Q_A} + \frac{0.489}{t}$ $+ 0.143\sqrt{\mu} - 0.143\sqrt{\rho} + 0.034N_d\rho$ $+ \frac{0.237\rho}{\chi_B} - 0.143\sqrt{N_d}\rho$ $\text{Final: } 0.720\frac{1}{t} + \frac{0.509}{Q_A} + 0.468\sqrt{Q_A + \rho} - 0.169\sqrt{\rho}$ $+ 0.238\rho\frac{1}{\chi_B} + 0.035N_d\rho - 0.148\sqrt{N_d}\rho$ $K_{\text{eff}}: 7$	0.091	0.099	0.061	0.061	68.618
5	$\text{Raw: } 0.288 + \frac{0.119}{R_A} + \frac{1.063}{t} + 0.222t$ $+ \frac{0.335\rho}{N_d} + 0.014N_d\rho - 0.068\chi_B\rho$	0.091	0.107	0.061	0.072	94.344

Continued on next page

Table 10: **HIERBOSSS** results across 5 independent 90/10 train/test splits of the oxide perovskite dataset. **Raw** descriptor correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** descriptor after post-MCMC symbolic model refinement. The descriptors correspond to the top-ranked by JMP. Train and test RMSE and MAE are computed on the corresponding 90% training and 10% held-out test sets, respectively. **Final** descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity. Effective symbolic forest size denoted by K_{eff} (*continued*).

Split	Learned expression	Train RMSE (eV)	Test RMSE (eV)	Train MAE (eV)	Test MAE (eV)	Runtime (s)
	Final: $0.371t + 1.288\frac{1}{t} + \frac{0.334\rho}{N_d}$ $+ 0.014N_d\rho - 0.069\chi_B\rho$ $K_{\text{eff}}: 5$	0.091	0.108	0.061	0.072	—

E.2 Top Descriptors Ranked by JMP

Table 11: Top $r = 10$ descriptors ranked by JMP learned by **HIERBOSSS** across the 5 independent train/test splits of the oxide perovskite dataset. **Raw** descriptor correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** descriptor after post-MCMC symbolic model refinement. Test RMSE and MAE are computed on the held-out 10% test set. **Final** descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity. Effective symbolic forest size denoted by K_{eff} .

Rank	Learned expression	Test RMSE (eV)	Test MAE (eV)	log JMP
<i>Split 1</i>				
1-7	Raw: $0.329 + 0.289\sqrt{\mu} + \frac{0.289}{\mu} - 0.004\rho^2$ $- \frac{0.674}{\sqrt{\chi_A} - \sqrt{\rho}} - 0.437\sqrt{\rho} + 0.006N_d\rho$ $+ \frac{0.310\rho}{\chi_B} + \frac{0.164\mu}{\rho}$ Final: $\frac{0.429}{\mu} - 0.004\rho^2 - \frac{0.811}{\sqrt{\chi_A} - \sqrt{\rho}}$ $- 0.454\sqrt{\rho} + 0.006N_d\rho + 0.339\rho\frac{1}{\chi_B} + \frac{0.206\mu}{\rho}$ $K_{\text{eff}}: 7$	0.084	0.057	1658.755
8	Raw: $0.363 + 0.265\mu + \frac{0.303}{\mu} - 0.004\rho^2$ $- \frac{0.679}{\sqrt{\chi_A} - \sqrt{\rho}} - 0.437\sqrt{\rho} + 0.006N_d\rho$ $+ \frac{0.311\rho}{\chi_B} + \frac{0.166\mu}{\rho}$ Final: $\frac{0.429}{\mu} - 0.004\rho^2 - \frac{0.811}{\sqrt{\chi_A} - \sqrt{\rho}}$ $- 0.454\sqrt{\rho} + 0.006N_d\rho + 0.339\rho\frac{1}{\chi_B} + \frac{0.206\mu}{\rho}$ $K_{\text{eff}}: 7$	0.084	0.057	1658.626
9-10	Raw: $0.469 + \frac{0.320}{\mu} + 0.430\sqrt{\mu} - 0.004\rho^2$ $- \frac{0.490}{\chi_A - \sqrt{\rho}} - 0.509\sqrt{\rho} + 0.006N_d\rho$ $+ \frac{0.098\mu}{\rho} + \frac{0.308\rho}{\chi_B}$	0.084	0.057	1657.536

Continued on next page

Table 11: Top $r = 10$ descriptors ranked by JMP learned by **HIERBOSSS** across the 5 independent train/test splits of the oxide perovskite dataset. *Raw* descriptor correspond to the symbolic forest learned by **HIERBOSSS** followed by the *Final* descriptor after post-MCMC symbolic model refinement. Test RMSE and MAE are computed on the held-out 10% test set. *Final* descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity. Effective symbolic forest size denoted by K_{eff} (*continued*).

Rank	Learned expression	Test RMSE (eV)	Test MAE (eV)	log JMP
	<i>Final:</i> $0.719\sqrt{\mu} + \frac{0.421}{\mu} - \frac{0.524}{\chi_A - \sqrt{\rho}}$ $- 0.004\rho^2 - 0.520\sqrt{\rho} + 0.006N_d\rho$ $+ 0.318\rho\frac{1}{\chi_B} + \frac{0.108\mu}{\rho}$ $K_{\text{eff}}: 8$	0.084	0.057	—
<i>Split 2</i>				
1-3	<i>Raw:</i> $0.297 + 0.236\sqrt{Q_A} + 0.357\sqrt{R_A} + 0.293\sqrt{\mu}$ $+ 0.335\sqrt{\chi_A} + \frac{0.538\rho}{N_d} + 0.097\sqrt{N_d}\rho$ $- 0.068\chi_B\rho - 0.076R_A\rho - 0.076\chi_A\rho$ <i>Final:</i> $1.158 + 0.292\sqrt{Q_A} + \frac{0.530\rho}{N_d} + 0.095\sqrt{N_d}\rho$ $- 0.068\chi_B\rho - 0.073R_A\rho - 0.073\chi_A\rho$ $K_{\text{eff}}: 5$	0.086	0.064	1660.707
4	<i>Raw:</i> $0.131 + \frac{0.262}{t} + 0.471\chi_A + 0.296\sqrt{\chi_A}$ $+ 0.322R_A + \frac{0.534\rho}{N_d} + 0.097\sqrt{N_d}\rho$ $- 0.071\chi_B\rho - 0.074R_A\rho - 0.074\chi_A\rho$ <i>Final:</i> $0.496R_A + 0.896\chi_A + \frac{0.596\rho}{N_d}$ $+ 0.108\sqrt{N_d}\rho - 0.070\chi_B\rho$ $- 0.089R_A\rho - 0.089\chi_A\rho$ $K_{\text{eff}}: 6$	0.086	0.063	1660.345
5	<i>Raw:</i> $0.376 + 0.383\chi_A + 0.263R_A + 0.333\sqrt{\mu}$ $+ 0.169\sqrt{Q_A} + \frac{0.558\rho}{N_d} + 0.100\sqrt{N_d}\rho$ $- 0.068\chi_B\rho - 0.081R_A\rho - 0.081\chi_A\rho$ <i>Final:</i> $0.496R_A + 0.896\chi_A + \frac{0.596\rho}{N_d}$ $+ 0.108\sqrt{N_d}\rho - 0.070\chi_B\rho$ $- 0.089R_A\rho - 0.089\chi_A\rho$ $K_{\text{eff}}: 6$	0.087	0.063	1660.010
6	<i>Raw:</i> $0.300 + 0.248\sqrt{R_A} + 0.272\sqrt{\chi_B} + 0.263\frac{1}{\chi_B}$ $+ 0.325\sqrt{Q_A} + \frac{0.542\rho}{N_d} + 0.098\sqrt{N_d}\rho$ $- 0.070\chi_B\rho - 0.076R_A\rho - 0.076\chi_A\rho$ <i>Final:</i> $1.158 + 0.292\sqrt{Q_A} + \frac{0.530\rho}{N_d} + 0.095\sqrt{N_d}\rho$ $- 0.068\chi_B\rho - 0.073R_A\rho - 0.073\chi_A\rho$ $K_{\text{eff}}: 5$	0.087	0.064	1659.242

Continued on next page

Table 11: Top $r = 10$ descriptors ranked by JMP learned by **HIERBOSSS** across the 5 independent train/test splits of the oxide perovskite dataset. **Raw** descriptor correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** descriptor after post-MCMC symbolic model refinement. Test RMSE and MAE are computed on the held-out 10% test set. **Final** descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity. Effective symbolic forest size denoted by K_{eff} (*continued*).

Rank	Learned expression	Test RMSE (eV)	Test MAE (eV)	log JMP
7–8	Raw: $0.321 + 0.276\sqrt{t} + 0.280\sqrt{\chi_B} + \frac{0.290}{\chi_B}$ $+ 0.305\sqrt{Q_A} + \frac{0.532\rho}{N_d} + 0.096\sqrt{N_d}\rho$ $- 0.070\chi_B\rho - 0.073R_A\rho - 0.073\chi_A\rho$	0.086	0.064	1659.127
	Final: $1.158 + 0.292\sqrt{Q_A} + \frac{0.530\rho}{N_d} + 0.095\sqrt{N_d}\rho$ $- 0.068\chi_B\rho - 0.073R_A\rho - 0.073\chi_A\rho$	0.087	0.064	—
	K_{eff}: 5			
9–10	Raw: $0.316 + \frac{1.016}{t} + \frac{0.070}{\chi_B} + 0.189\mu + 0.209\sqrt{t}$ $+ \frac{0.315\rho}{N_d} + 0.013N_d\rho - 0.063\chi_B\rho$	0.086	0.063	1659.092
	Final: $1.181\frac{1}{t} + 0.475\sqrt{t} + \frac{0.327\rho}{N_d}$ $+ 0.014N_d\rho - 0.066\chi_B\rho$	0.086	0.063	—
	K_{eff}: 5			
Split 3				
1–3	Raw: $0.219 + 0.357\mu + 0.352\sqrt{R_A} + 0.294\chi_A + \frac{0.338}{t}$ $+ 0.343\sqrt{\mu} + 0.006N_d\rho + \frac{0.290\rho}{\chi_B^2} - 0.257\mu\rho$	0.093	0.059	1663.177
	Final: $2.103\sqrt{\mu} + 0.282\chi_A + 0.254\rho\frac{1}{\chi_B}\frac{1}{\chi_B}$ $+ 0.005N_d\rho - 0.214\mu\rho$	0.094	0.060	—
	K_{eff}: 5			
4–5	Raw: $0.160 + 0.588\sqrt{\chi_A} + 0.306\sqrt{R_A} + \frac{0.341}{t}$ $+ 0.446\mu + 0.006N_d\rho + \frac{0.295\rho}{\chi_B^2} - 0.262\mu\rho$	0.094	0.059	1662.851
	Final: $0.753 + 0.645\mu + 0.608\sqrt{\chi_A}$ $+ 0.006N_d\rho + 0.294\rho\frac{1}{\chi_B}\frac{1}{\chi_B} - 0.260\mu\rho$	0.094	0.059	—
	K_{eff}: 5			
6–7	Raw: $0.325 + 0.383\sqrt{\chi_A} + \frac{0.720}{\chi_B} + 0.071N_d$ $+ \frac{0.765}{N_d} - 0.088\rho + \frac{0.188\rho}{\chi_B}$	0.095	0.062	1662.809
	Final: $0.565\sqrt{\chi_A} + \frac{1.030}{N_d} + 0.077N_d$ $+ 0.820\frac{1}{\chi_B} - 0.080\rho + 0.173\rho\frac{1}{\chi_B}$	0.096	0.063	—
	K_{eff}: 6			

Continued on next page

Table 11: Top $r = 10$ descriptors ranked by JMP learned by **HIERBOSSS** across the 5 independent train/test splits of the oxide perovskite dataset. **Raw** descriptor correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** descriptor after post-MCMC symbolic model refinement. Test RMSE and MAE are computed on the held-out 10% test set. **Final** descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity. Effective symbolic forest size denoted by K_{eff} (*continued*).

Rank	Learned expression	Test RMSE (eV)	Test MAE (eV)	log JMP
8–9	Raw: $0.217 + \frac{0.359}{t} + 0.383\sqrt{R_A} + 0.209\sqrt{\chi_A}$ $+ 0.430\mu + 0.217\chi_A + 0.006N_d\rho$ $+ \frac{0.294\rho}{\chi_B^2} - 0.262\mu\rho$	0.093	0.059	1662.647
	Final: $1.034 + 0.310\chi_A + 0.684\mu + 0.006N_d\rho$ $+ 0.291\rho \frac{1}{\chi_B} \frac{1}{\chi_B} - 0.257\mu\rho$			
	$K_{\text{eff}}: 5$			
10	Raw: $0.270 + 0.069N_d + 0.249\sqrt{\chi_A} + \frac{0.506}{\chi_B}$ $+ \frac{0.301}{t} + \frac{0.886}{N_d} - 0.090\rho + \frac{0.192\rho}{\chi_B}$	0.096	0.062	1662.538
	Final: $0.069N_d + \frac{1.303}{N_d} + 1.038 \frac{1}{t}$ $- 0.094\rho + 0.200\rho \frac{1}{\chi_B}$			
	$K_{\text{eff}}: 5$			
Split 4				
1	Raw: $0.299 + 0.401\sqrt{Q_A + \rho} + \frac{0.356}{Q_A} + \frac{0.489}{t}$ $+ 0.143\sqrt{\mu} - 0.143\sqrt{\rho} + 0.034N_d\rho$ $+ \frac{0.237\rho}{\chi_B} - 0.143\sqrt{N_d\rho}$	0.099	0.061	1674.761
	Final: $0.720 \frac{1}{t} + \frac{0.509}{Q_A} + 0.468\sqrt{Q_A + \rho} - 0.169\sqrt{\rho}$ $+ 0.238\rho \frac{1}{\chi_B} + 0.035N_d\rho - 0.148\sqrt{N_d\rho}$			
	$K_{\text{eff}}: 7$			
2–3	Raw: $0.329 + \frac{0.506}{t} + \frac{0.364}{Q_A} + 0.406\sqrt{Q_A + \rho}$ $+ 0.081\mu - 0.145\sqrt{\rho} + 0.034N_d\rho$ $+ \frac{0.237\rho}{\chi_B} - 0.143\sqrt{N_d\rho}$	0.099	0.061	1674.644
	Final: $0.720 \frac{1}{t} + \frac{0.509}{Q_A} + 0.468\sqrt{Q_A + \rho} - 0.169\sqrt{\rho}$ $+ 0.238\rho \frac{1}{\chi_B} + 0.035N_d\rho - 0.148\sqrt{N_d\rho}$			
	$K_{\text{eff}}: 7$			
4–5	Raw: $0.318 + \frac{0.399}{Q_A} + \frac{0.486}{t} + 0.047\chi_A$ $+ 0.407\sqrt{Q_A + \rho} - 0.145\sqrt{\rho} + 0.034N_d\rho$ $+ \frac{0.239\rho}{\chi_B} - 0.144\sqrt{N_d\rho}$	0.099	0.061	1674.184
	Final: $0.720 \frac{1}{t} + \frac{0.509}{Q_A} + 0.468\sqrt{Q_A + \rho} - 0.169\sqrt{\rho}$ $+ 0.238\rho \frac{1}{\chi_B} + 0.035N_d\rho - 0.148\sqrt{N_d\rho}$			
	$K_{\text{eff}}: 7$			

Continued on next page

Table 11: Top $r = 10$ descriptors ranked by JMP learned by **HIERBOSSS** across the 5 independent train/test splits of the oxide perovskite dataset. **Raw** descriptor correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** descriptor after post-MCMC symbolic model refinement. Test RMSE and MAE are computed on the held-out 10% test set. **Final** descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity. Effective symbolic forest size denoted by K_{eff} (*continued*).

Rank	Learned expression	Test RMSE (eV)	Test MAE (eV)	log JMP
6–10	Raw: $0.301 + 0.405\sqrt{Q_A} + \rho + \frac{0.530}{t} + \frac{0.350}{Q_A}$ $+ 0.037\sqrt{R_A} - 0.144\sqrt{\rho} + \frac{0.238\rho}{\chi_B}$ $+ 0.034N_d\rho - 0.144\sqrt{N_d\rho}$ Final: $0.720\frac{1}{t} + \frac{0.509}{Q_A} + 0.468\sqrt{Q_A} + \rho - 0.169\sqrt{\rho}$ $+ 0.238\rho\frac{1}{\chi_B} + 0.035N_d\rho - 0.148\sqrt{N_d\rho}$ $K_{\text{eff}}: 7$	0.099	0.061	1674.116
Split 5				
1	Raw: $0.288 + \frac{0.119}{R_A} + \frac{1.063}{t} + 0.222t$ $+ \frac{0.335\rho}{N_d} + 0.014N_d\rho - 0.068\chi_B\rho$ Final: $0.371t + 1.288\frac{1}{t} + \frac{0.334\rho}{N_d}$ $+ 0.014N_d\rho - 0.069\chi_B\rho$ $K_{\text{eff}}: 5$	0.107	0.072	1685.919
2	Raw: $0.254 + 0.262\sqrt{\mu} + 0.247\sqrt{t} + \frac{0.991}{t}$ $+ \frac{0.329\rho}{N_d} + 0.014N_d\rho - 0.067\chi_B\rho$ Final: $0.506\sqrt{t} + 1.153\frac{1}{t} + \frac{0.335\rho}{N_d}$ $+ 0.014N_d\rho - 0.069\chi_B\rho$ $K_{\text{eff}}: 5$	0.108	0.072	1685.775
3	Raw: $0.288 + \frac{1.126}{t} + 0.190t + \frac{0.103}{\chi_B}$ $+ \frac{0.324\rho}{N_d} + 0.014N_d\rho - 0.066\chi_B\rho$ Final: $0.371t + 1.288\frac{1}{t} + \frac{0.334\rho}{N_d}$ $+ 0.014N_d\rho - 0.069\chi_B\rho$ $K_{\text{eff}}: 5$	0.108	0.072	1685.605
4–6	Raw: $0.342 + 0.386\mu + 0.159t + \frac{0.999}{t}$ $+ \frac{0.324\rho}{N_d} + 0.014N_d\rho - 0.066\chi_B\rho$ Final: $0.371t + 1.288\frac{1}{t} + \frac{0.334\rho}{N_d}$ $+ 0.014N_d\rho - 0.069\chi_B\rho$ $K_{\text{eff}}: 5$	0.108	0.072	1685.392
7	Raw: $0.257 + 0.098t + \frac{1.232}{t} + \frac{0.082}{\chi_A}$ $+ \frac{0.332\rho}{N_d} + 0.014N_d\rho - 0.068\chi_B\rho$	0.107	0.072	1685.288

Continued on next page

Table 11: Top $r = 10$ descriptors ranked by JMP learned by **HIERBOSSS** across the 5 independent train/test splits of the oxide perovskite dataset. **Raw** descriptor correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** descriptor after post-MCMC symbolic model refinement. Test RMSE and MAE are computed on the held-out 10% test set. **Final** descriptors highlighted in — were identified by Weng et al. [2020][Table 2] as particularly important for OER activity. Effective symbolic forest size denoted by K_{eff} (*continued*).

Rank	Learned expression	Test RMSE (eV)	Test MAE (eV)	log JMP
	Final: $1.454 \frac{1}{t} + 0.228 \frac{1}{\chi_A} + \frac{0.331\rho}{N_d}$ $+ 0.014N_d\rho - 0.067\chi_B\rho$ K_{eff}: 5	0.107	0.072	—
8–9	Raw: $0.276 + 0.242\sqrt{R_A} + \frac{0.267}{R_A} + \frac{0.841}{t}$ $+ \frac{0.116}{\chi_B} + \frac{0.325\rho}{N_d} + 0.014N_d\rho$ $- 0.067\chi_B\rho$ Final: $1.230 \frac{1}{t} + 0.370\sqrt{R_A} + \frac{0.328\rho}{N_d}$ $+ 0.014N_d\rho - 0.067\chi_B\rho$ K_{eff}: 5	0.108	0.072	1685.248
10	Raw: $0.415 + \frac{0.694}{t} + 0.756\sqrt{\mu} + 0.158\mu$ $+ \frac{0.317\rho}{N_d} + 0.013N_d\rho - 0.064\chi_B\rho$ Final: $0.736 + 0.924 \frac{1}{t} + \frac{0.334\rho}{N_d}$ $+ 0.014N_d\rho - 0.068\chi_B\rho$ K_{eff}: 4	0.108	0.072	1685.105
	Final: $0.736 + 0.924 \frac{1}{t} + \frac{0.334\rho}{N_d}$ $+ 0.014N_d\rho - 0.068\chi_B\rho$ K_{eff}: 4	0.107	0.072	—

E.3 Prediction Plots of V_{RHE} for HierBOSSS

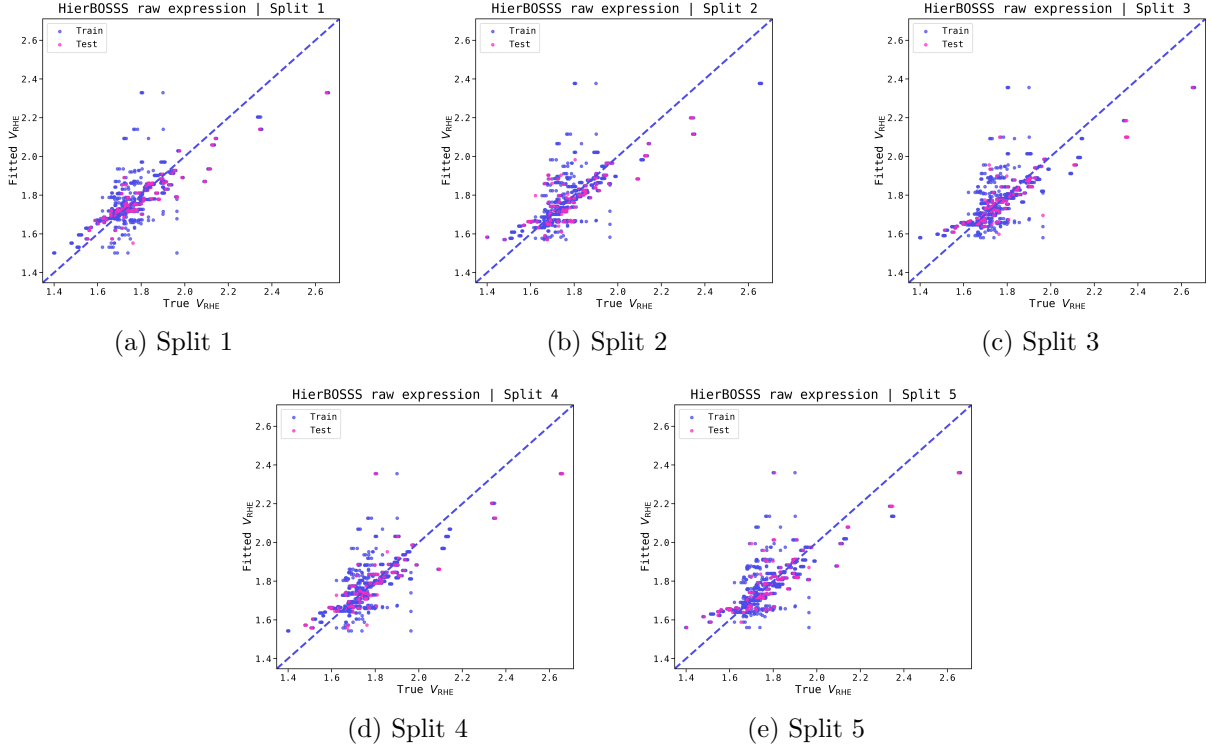


Figure 14: True versus fitted V_{RHE} values obtained by *Raw* descriptors of **HIERBOSSS** across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

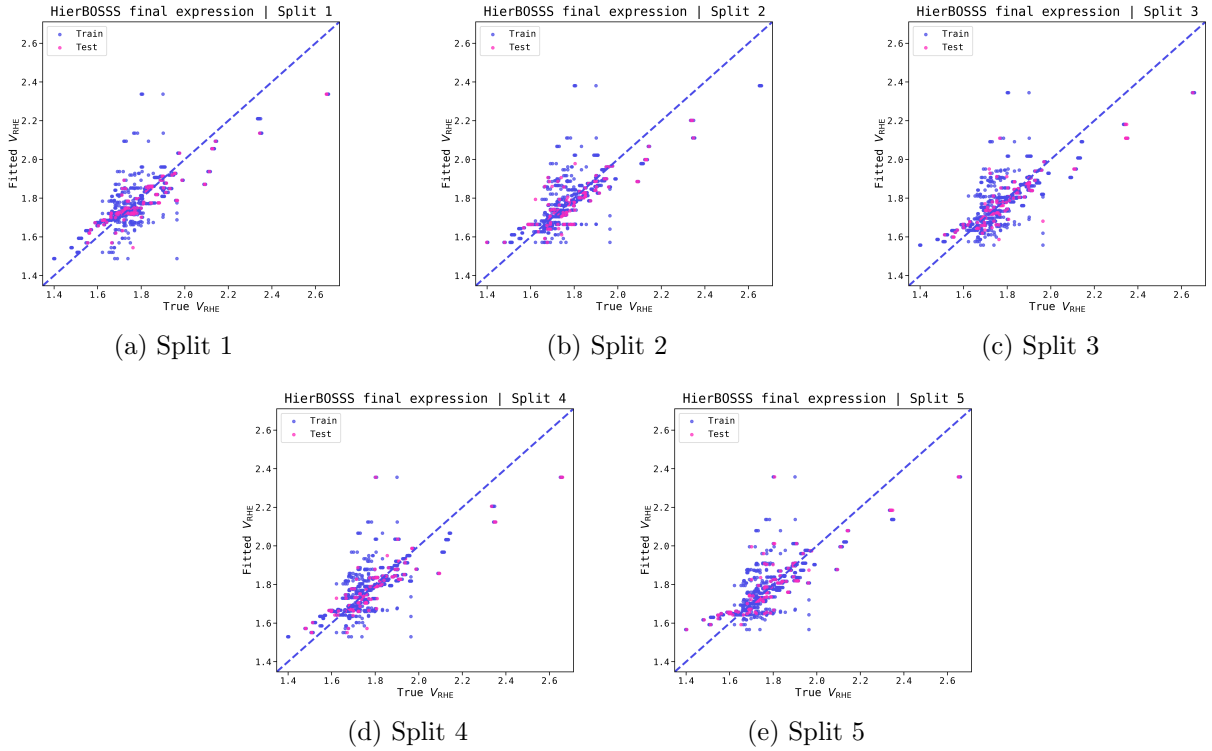


Figure 15: True versus fitted V_{RHE} values obtained by *Final* descriptors of **HIERBOSSS** across 5 independent 90/10 train/test splits of the oxide perovskite dataset.

E.4 Posterior Diagnostics for **HierBOSSS**

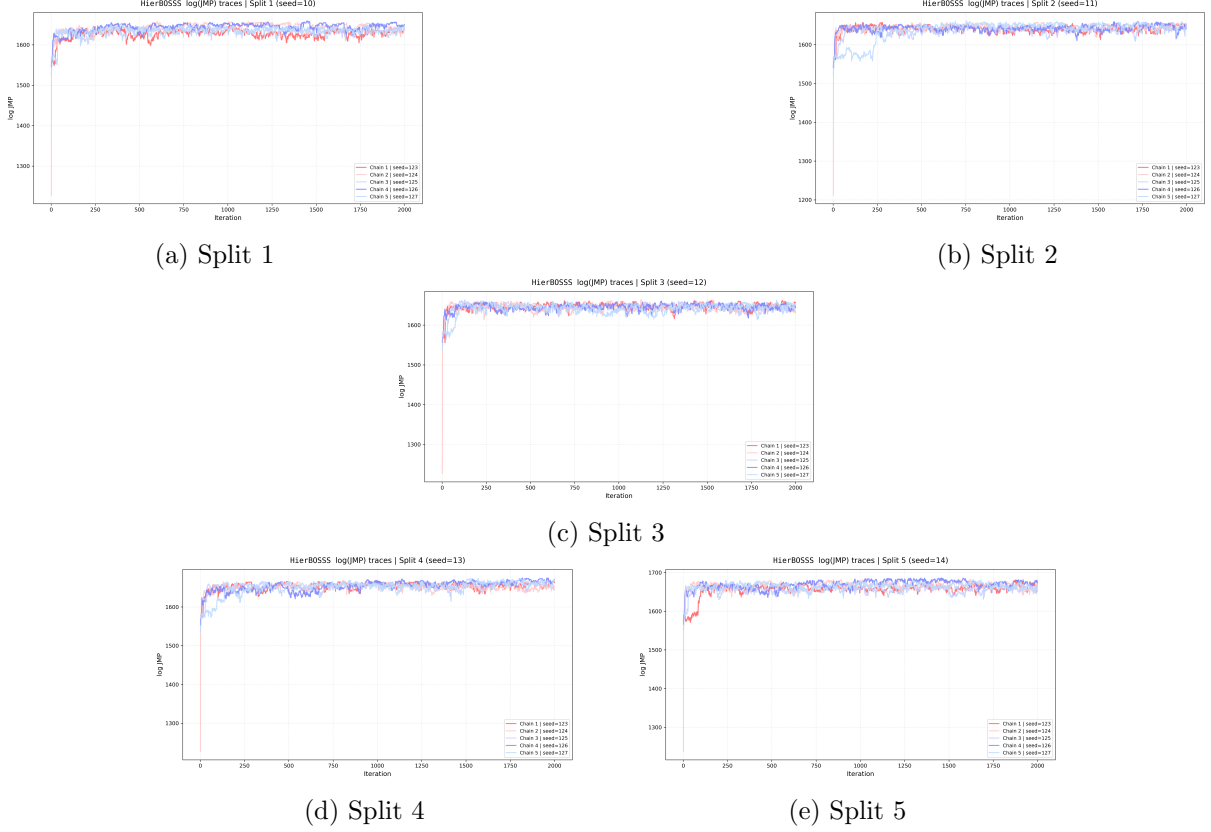


Figure 16: Trace plots of **HierBOSSS** across the 5 splits of the oxide perovskite dataset.

E.5 Runtime Comparison of **HierBOSSS** with the Bayesian SR Methods

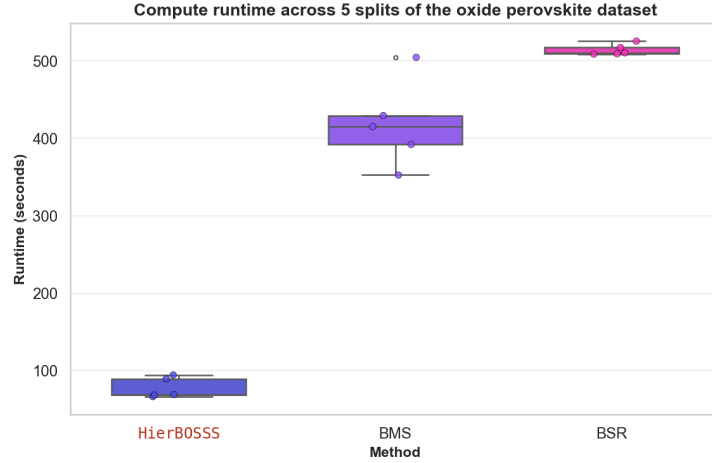


Figure 17: Compute runtimes of **HierBOSSS**, BMS (5 parallel MCMC chains), and BSR (5 restarts) across 5 splits of the oxide perovskite dataset. Implementation details of **HierBOSSS** and the Bayesian competitors follow from [Section B.1](#), [B.8](#), and [B.9](#).

F Accuracy-Complexity Trade-off of Learned Descriptors

We compare the predictive accuracy and symbolic complexity of the descriptors learned by **HierBOSSS** and competing SR methods on the oxide perovskite dataset. In light of our observations from the experiments on Feynman equations, predictive accuracy alone does not fully characterize the scientific interpretability of a learned descriptor, since a low-error expression may still contain excessive nesting, redundant transformations, or many constants and operators. We therefore evaluate held-out test MAE jointly with symbolic model size [Imai Aldeia et al., 2025, La Cava et al., 2021] across the 5 independent 90/10 train/test splits of the dataset. Experimental configurations for **HierBOSSS** and the competing methods are kept the same as those described in Section B.

For each learned symbolic expression representing a descriptor, symbolic model size is obtained similarly as in the case of Feynman equations. For **HierBOSSS**, complexity is computed from the *Final* descriptor expression after post-MCMC symbolic model refinement; for each competing method, it is computed from the descriptor expression returned by the corresponding implementation.

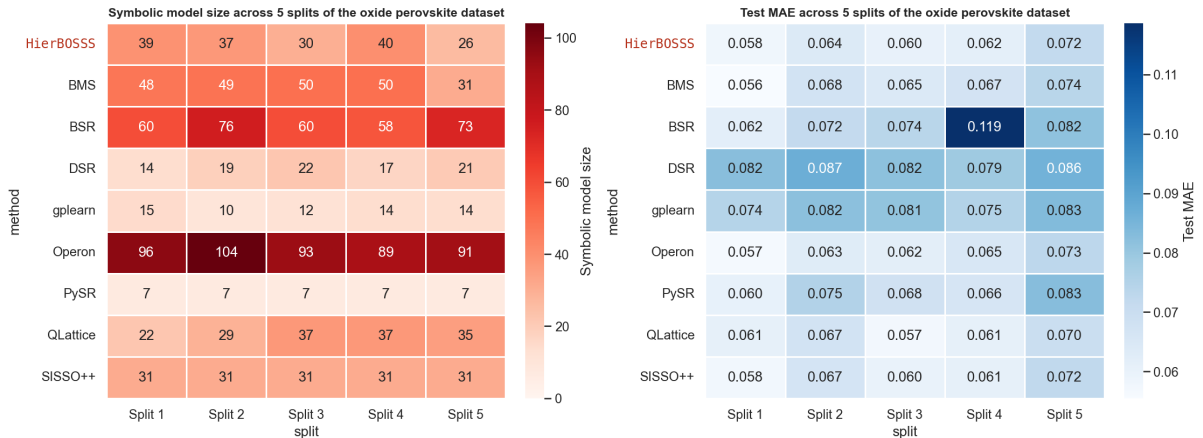


Figure 18: Symbolic model size and test MAE of learned descriptor expressions across 5 independent train/test splits of the oxide perovskite dataset.

Figure 18 shows that low predictive error does not necessarily imply symbolic parsimony. **operon** and **BSR** often attain competitive test MAE through substantially larger descriptor expressions, while **PySR** and **gplearn** return very compact symbolic models but generally exhibit weaker predictive performance. **BMS**, **QLattice**, and **SISSO++** occupy intermediate regimes, with moderate-to-large descriptor expressions and competitive accuracy on selected data splits.

The joint accuracy-complexity plot in Figure 19 provides a clearer view of this trade-off. Across the 5 data splits, **HierBOSSS** consistently attains low test MAE with substantially smaller symbolic models than several high-performing competitors. Its *Final* descriptor expressions contain only 26-40 nodes, while descriptor expressions from several competitors often require 50-100 nodes to achieve comparable predictive accuracy. Moreover, the learned expressions repeatedly recover descriptors such as t , $1/t$, μ , $\sqrt{\mu}$, $\sqrt{Q_A}$, χ_A , $\sqrt{\chi_A}$, χ_B , and $1/\chi_B$ which were also identified by Weng et al. [2020][Table 2] as scientifically important for OER activity. This favorable behavior is consistent with the Feynman experiments and reflects the combined effect of the Occam’s

razor-supported depth regularizing symbolic tree prior [Jefferys and Berger, 1992] and the post-MCMC symbolic model refinement applied to the Occam’s window-based model selection using $\text{JMP}(\mathcal{T})$ [Madigan and Raftery, 1994], which preserves high-posterior, scientifically meaningful descriptor structures while removing redundant symbolic components. Overall, **HierBOSSS** achieves a strong balance among predictive accuracy, symbolic parsimony, and scientific interpretability.

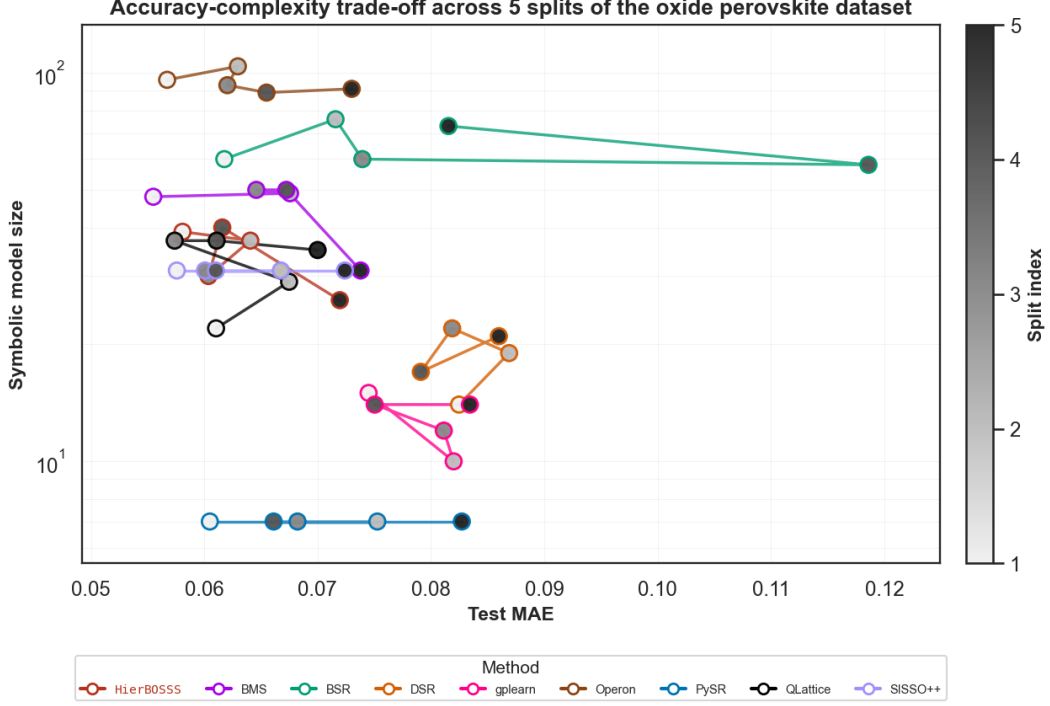


Figure 19: Accuracy-complexity trade-off across 5 independent train/test splits of the oxide perovskite dataset. Each point represents one method on one split, with test MAE on the horizontal axis and symbolic model size on the vertical axis. Point shading indicates the split index.

References

- Christopher J. Bartel, Christopher Sutton, Bryan R. Goldsmith, Runhai Ouyang, Charles B. Musgrave, Luca M. Ghiringhelli, and Matthias Scheffler. New tolerance factor to predict the stability of perovskite oxides and halides. *Science Advances*, 5(2):eaav0693, 2019.
- Kevin Broløs, Meera Vieira René, Machado, Chris Cave, Jaan Kasak, Valdemar Stentoft-Hansen, Victor Galindo Batanero, Tom Jelen, and Casper Wilstrup. An approach to symbolic regression using feyn. *arXiv:2104.05417*, 2021.
- Bogdan Burlacu, Gabriel Kronberger, and Michael Kommenda. Operon C++: an efficient genetic programming framework for symbolic regression. In *Proceedings of the 2020 Genetic and Evolutionary Computation Conference Companion*, 2020.
- Miles Cranmer. Interpretable Machine Learning for Science with PySR and SymbolicRegression.jl. *arXiv:2305.01582*, 2023.
- Lucas Foppa, Luca M. Ghiringhelli, Frank Girgsdies, Maike Hashagen, Pierre Kube, Michael Hävecker, Spencer J. Carey, Andrey Tarasov, Peter Kraus, Frank Rosowski, Robert Schlögl, Annette Trunschke, and Matthias Scheffler. Materials Genes of Heterogeneous Catalysis from Clean Experiments and Artificial Intelligence. *MRS Bulletin*, 46:1016–1026, 2021.
- Luca M. Ghiringhelli, Jan Vybiral, Sergey V. Levchenko, Claudia Draxl, and Matthias Scheffler. Big Data of Materials Science: Critical Role of the Descriptor. *Phys. Rev. Lett.*, 114:105503, 2015.
- Luca M Ghiringhelli, Jan Vybiral, Emre Ahmetcik, Runhai Ouyang, Sergey V Levchenko, Claudia Draxl, and Matthias Scheffler. Learning physical descriptors for materials science by compressed sensing. *New Journal of Physics*, 19(2):023017, 2017.
- Roger Guimerà, Ignasi Reichardt, Antoni Aguilar-Mogas, Francesco A Massucci, Manuel Miranda, Jordi Pallarès, and Marta Sales-Pardo. A Bayesian machine scientist to aid in the solution of challenging scientific problems. *Science advances*, 6(5):eaav6971, 2020.
- Zhong-Kang Han, Debalaya Sarker, Runhai Ouyang, Aliaksei Mazheika, Yi Gao, and Sergey V Levchenko. Single-atom alloy catalysts designed by first-principles calculations and artificial intelligence. *Nature Communications*, 12(1):1833, 2021.
- Jonathan Hwang, Reshma R. Rao, Livia Giordano, Yu Katayama, Yang Yu, and Yang Shao-Horn. Perovskites in catalysis and electrocatalysis. *Science*, 358(6364):751–756, 2017.
- Guilherme Seidyo Imai Aldeia, Hengzhe Zhang, Geoffrey Bomarito, Miles Cranmer, Alcides Fonseca, Bogdan Burlacu, William G. La Cava, and Fabrício Olivetti de França. Call for Action: towards the next generation of symbolic regression benchmark. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, 2025.
- William H. Jefferys and James O. Berger. Ockham’s Razor and Bayesian Analysis. *American Scientist*, 80(1):64–72, 1992.
- Ying Jin, Weilin Fu, Jian Kang, Jiadong Guo, and Jian Guo. Bayesian Symbolic Regression. *arXiv:1910.08892*, 2020.

- William La Cava, Patryk Orzechowski, Bogdan Burlacu, Fabricio de Franca, Marco Virgolin, Ying Jin, Michael Kommenda, and Jason Moore. Contemporary Symbolic Regression Methods and their Relative Performance. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.
- David Madigan and Adrian E Raftery. Model Selection and Accounting for Model Uncertainty in Graphical Models using Occam’s Window. *Journal of the American Statistical Association*, 89(428):1535–1546, 1994.
- Aaron Meurer, Christopher P. Smith, Mateusz Paprocki, Ondrej Certik, Sergey B. Kirpichev, Matthew Rocklin, Amit Kumar, Sergiu Ivanov, Jason K. Moore, Sartaj Singh, Thilina Rathnayake, Sean Vig, Brian E. Granger, Richard P. Muller, Francesco Bonazzi, Harsh Gupta, Shivam Vats, Fredrik Johansson, Fabian Pedregosa, Matthew J. Curry, Andy R. Terrel, Stjepan Roucka, Ashutosh Saboo, Isuru Fernando, Sumith Kulal, Robert Cimrman, and Anthony Scopatz. SymPy: symbolic computing in Python. *PeerJ Computer Science*, 3:e103, 2017.
- Sebastian Musslick, Benjamin Andrew, Chad C. Williams, Joshua T. S. Hewson, Sida Li, Ioana Marinescu, Marina Dubova, George T. Dang, Younes Strittmatter, and John G. Holland. AutoRA: Automated Research Assistant for Closed-Loop Empirical Research. *Journal of Open Source Software*, 9(104):6839, 2024.
- Runhai Ouyang, Stefano Curtarolo, Emre Ahmetcik, Matthias Scheffler, and Luca M. Ghiringhelli. SISSO: A compressed-sensing method for identifying the best low-dimensional descriptor in an immensity of offered candidates. *Phys. Rev. Mater.*, 2:083802, 2018.
- Brenden K Petersen, Mikel Llandajueta Larma, Terrell N. Mundhenk, Claudio Prata Santiago, Soo Kyung Kim, and Joanne Taery Kim. Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients. In *International Conference on Learning Representations*, 2021.
- Thomas A. R. Purcell, Matthias Scheffler, Christian Carbogno, and Luca M. Ghiringhelli. SISSO++: A C++ implementation of the sure-independence screening and sparsifying operator approach. *Journal of Open Source Software*, 7(71):3960, 2022.
- Trevor Stephens. gplearn: Genetic Programming in Python, 2016. URL <https://github.com/trevorstephens/gplearn>.
- Jin Suntivich, Kevin J. May, Hubert A. Gasteiger, John B. Goodenough, and Yang Shao-Horn. A Perovskite Oxide Optimized for Oxygen Evolution Catalysis from Molecular Orbital Principles. *Science*, 334(6061):1383–1385, 2011.
- Guanjie Wang, , Erpeng Wang, Zefeng Li, Jian Zhou, and Zhimei Sun. Exploring the mathematical equations behind the materials science data using interpretable symbolic regression. *Interdisciplinary Materials*, 3(5):637–657, 2024.
- Baicheng Weng, Zhilong Song, Rilong Zhu, Qingyu Yan, Qingde Sun, Corey G. Grice, Yanfa Yan, and Wan-Jian Yin. Simple Descriptor Derived from Symbolic Regression Accelerating the Discovery of New Perovskite Catalysts. *Nature Communications*, 11(1):3513, 2020.